

Chapter 7 Series FACTS Controllers: GCSC, TSSC, TCSC and SSSC

Learning Objectives

After working through this chapter, the reader should be able to:

- derive the current, transmitted power, midpoint voltage and capacitor reactive power of a series-compensated line, and explain why series compensation is more effective per MVA than shunt compensation for raising transfer capability;
- quantify the influence of series compensation on voltage stability, transient stability and power oscillation damping, and compute the subsynchronous resonance frequencies it introduces;
- derive the capacitor voltage, fundamental reactance, RMS voltage and harmonics of the GTO-controlled series capacitor and exploit its duality with the TCR;
- explain the insertion and bypass rules of the thyristor-switched series capacitor and design stepped banks for voltage or impedance compensation;
- analyse the thyristor-controlled series capacitor with both the simplified parallel-LC model and the exact line-current-driven model, explain why they differ, and use the exact closed-form reactance law for design;
- derive the power-angle characteristic of the static synchronous series compensator, explain its immunity to series resonance, its power-reversal capability and its use with an energy store to compensate line resistance;
- compare the four controllers on range, response, harmonics, SSR behaviour, losses and cost, and place them in the current research landscape.

Table 7.1 summarizes the four controllers treated in this chapter, together with the fixed series capacitor from which they all derive.

Table 7.1 Series FACTS controllers at a glance.

Device	Power stage	Control principle	Output	Characteristic behaviour
Fixed series capacitor	Capacitor bank with MOV and bypass	None (switched by breaker)	Fixed $-jX_C$	Raises transfer; introduces SSR risk
GCSC	Capacitor + antiparallel GTO/IGCT valve	Turn-off delay γ from current peak	Continuous $0 \leq X \leq X_C$ (capacitive)	Dual of the TCR; harmonics generated
TSSC	Capacitor stages with thyristor bypass valves	Insert at current zero, bypass at voltage zero	Stepped capacitive	No harmonics; offset transients; SSR as a capacitor
TCSC	Capacitor in parallel with thyristor-controlled reactor	Conduction (firing) angle	Continuous capacitive boost and inductive vernier	Inductive at subsynchronous frequencies; most widely used
SSSC	VSC behind series transformer	Magnitude (and angle) of injected voltage	$\pm V_q$ independent of current	No series resonance; can reverse power; with storage, compensates R

7.1 Fundamentals of Series Compensation

7.1.1 Two ways of looking at a series capacitor

Consider the lossless two-machine system of Figure 7.1 with equal end voltages $V_s = V_r = V$, total line reactance X and a series capacitor of reactance X_C located at the centre of the line.

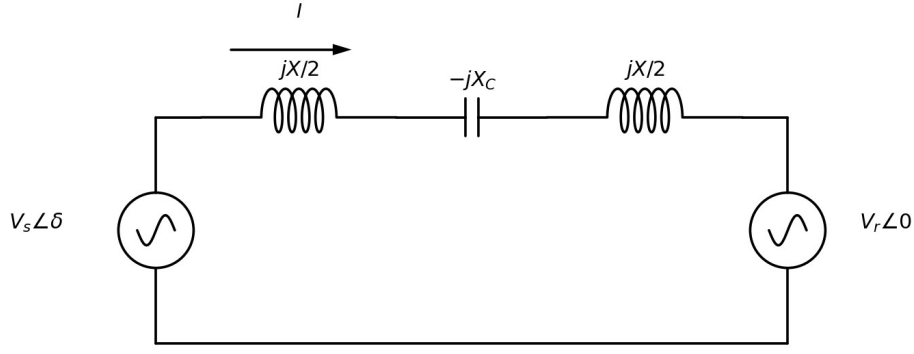


Figure 7.1. Two-machine system with a series capacitor at the centre of the line.

The conventional explanation is that the capacitor cancels part of the line reactance, so the line appears electrically shorter. Defining the *degree of series compensation*

$$k = \frac{X_C}{X}, 0 \leq k < 1 \quad (7.1)$$

where k is the degree of compensation and X_C the capacitor reactance, the effective reactance between the two ends is

$$X_{eff} = X - X_C = (1 - k)X \quad (7.2)$$

where X_{eff} is the net series reactance. The second explanation, which proves more useful once converters are involved, looks at voltages rather than impedances. To push a larger current through the fixed line reactance X , a larger voltage must appear across it. The capacitor supplies that extra voltage: at a given current it develops a voltage $-jX_C I$ that is opposite in phase to the drop jXI , so the voltage available across the line inductance is the sum of the end-voltage difference and the capacitor voltage. Figure 7.2 shows this. Once compensation is seen as a series *voltage* in quadrature with the line current, the physical nature of the element that produces it no longer matters, which is exactly the step from the capacitor to the SSSC.

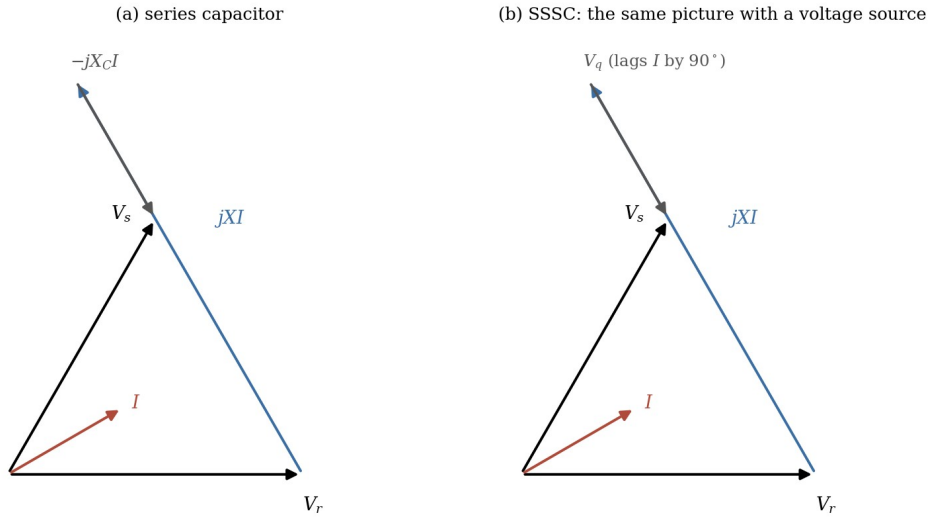


Figure 7.2. Phasor diagrams of (a) a series-capacitor-compensated line and (b) the same line with an SSSC producing the equivalent quadrature voltage.

7.1.2 Current, power and reactive power of the compensated line

With the end voltages $V \angle \delta/2$ and $V \angle -\delta/2$, the line current magnitude, the voltage at the electrical midpoint and the transmitted power are

$$I = \frac{2V}{(1-k)X} \sin \frac{\delta}{2}, V_m = V \cos \frac{\delta}{2}, P = \frac{V^2}{(1-k)X} \sin \delta \quad (7.3)$$

where I is the RMS line current, V_m the voltage at the electrical midpoint of a symmetrically compensated line and P the transmitted active power. The power-angle curve is simply scaled by $1/(1-k)$, so the maximum transmissible power rises to $V^2/[(1-k)X]$: 33% compensation raises it by half, 50% doubles it. The reactive power supplied by the capacitor is

$$Q_C = I^2 X_C = \frac{2V^2}{X} \cdot \frac{k}{(1-k)^2} (1 - \cos \delta) \quad (7.4)$$

where Q_C is the capacitor's reactive output. Figure 7.3 plots both quantities. The reactive power grows rapidly with k and with δ , because it scales with the square of a current that itself rises with compensation. This sets the economic limit of series compensation, together with the fault-current, insulation and SSR limits discussed below. Examples 7.13 to 7.15 work through these relations numerically.

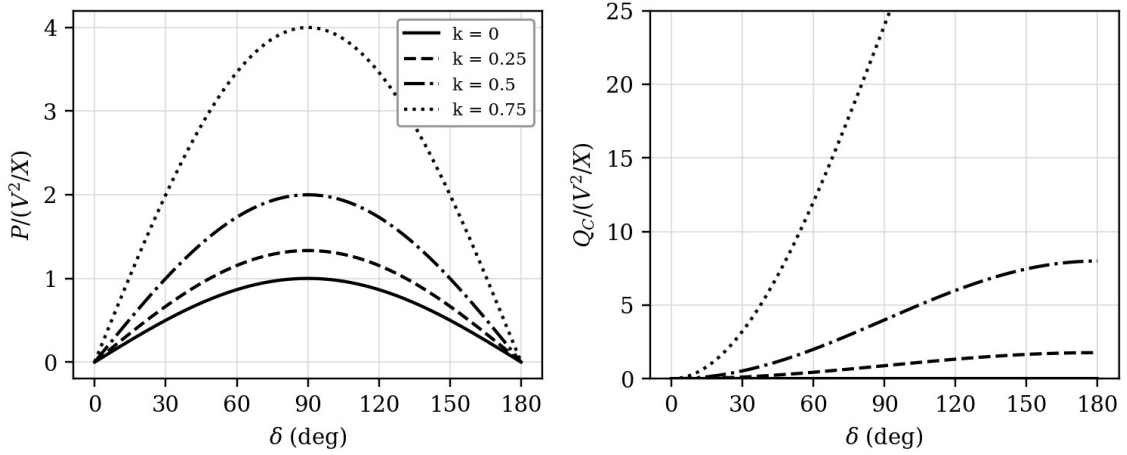


Figure 7.3. Transmitted power (left) and series-capacitor reactive power (right) versus angle for several degrees of compensation.

Why is series compensation more effective per MVA than shunt compensation for raising power transfer? A shunt compensator raises transfer only by supporting the voltage at an intermediate point; its MVA is spent holding a voltage magnitude. A series device acts on the transfer impedance itself, and its MVA, $I^2 X_C$, scales with the current that is actually being transmitted. For moderate angles, the series rating needed for a given increase in transmitted power is a fraction of the shunt rating that would produce the same increase. This is why long-distance AC corridors in Brazil, North America, Scandinavia and India are series compensated.

7.1.3 Voltage stability

For a radial line feeding a load, series compensation reduces the reactance through which the load is supplied, which raises the maximum loadability and pushes the nose of the P-V curve outward (Figure 7.4). The capacitor also self-regulates in a useful sense: its reactive output $I^2 X_C$ rises as the load current rises, precisely when the line's own reactive losses increase. For this reason series compensation is effective against voltage collapse on long radial supplies.

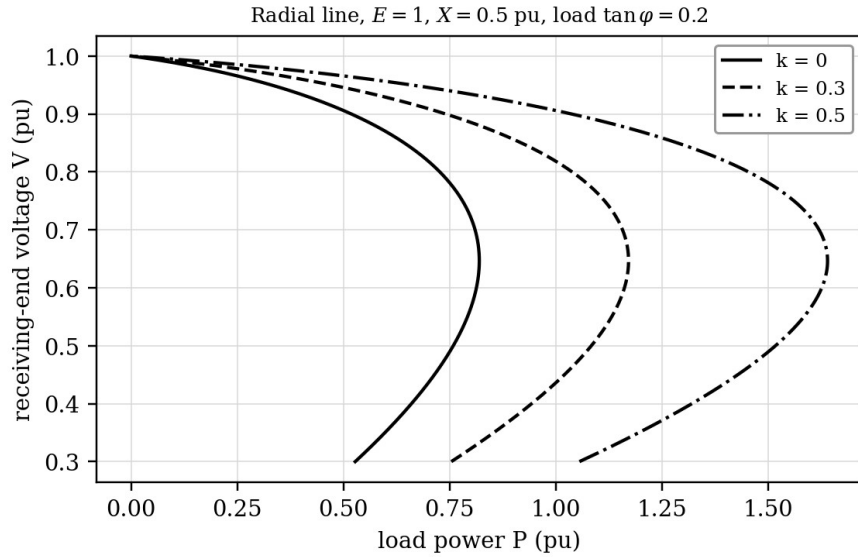


Figure 7.4. Receiving-end P-V curves of a radial line with 0, 30% and 50% series compensation.

Example 7.1. Effect of series compensation on steady-state transfer capability

A lossless 220 kV corridor is represented by a series reactance of $40 \, \Omega$ per phase. Both end voltages are 220 kV line-to-line and the transmission angle is 30° . Determine the transmitted power before compensation and with 40% series compensation.

Solution. With line-to-line voltage, the three-phase power of a lossless line is $P = V_{LL}^2 \sin \delta / X$. Without compensation, $P = 220^2 \times 0.5 / 40 = 605 \text{ MW}$. With $k = 0.40$, $X_{eff} = 40(1 - 0.40) = 24 \, \Omega$ and $P = 220^2 \times 0.5 / 24 = 1008.3 \text{ MW}$. The same angle now carries 66.7% more power, exactly the factor $1/(1 - k)$.

Note. The result shows the leverage of series compensation, not a permissible transfer: thermal limits, transient stability, voltage profile, protection duty and subsynchronous interaction must still be checked.

7.1.4 Transient stability and power oscillation damping

Raising the power-angle curve does two things in a transient. It lowers the pre-fault angle for the same mechanical power, and it enlarges the decelerating area available after clearing. Figure 7.5 illustrates the equal-area construction for an uncompensated line and for the same line with $k = 1/3$. For a fault that reduces the transfer capability to 0.4 pu, the critical clearing angle rises from about 91° to about 109° (Example 7.18). Controlled series compensation goes further. By switching to maximum compensation immediately after the fault, the controller raises the post-fault curve exactly when it matters, and then returns to the steady-state level once the first swing is over.

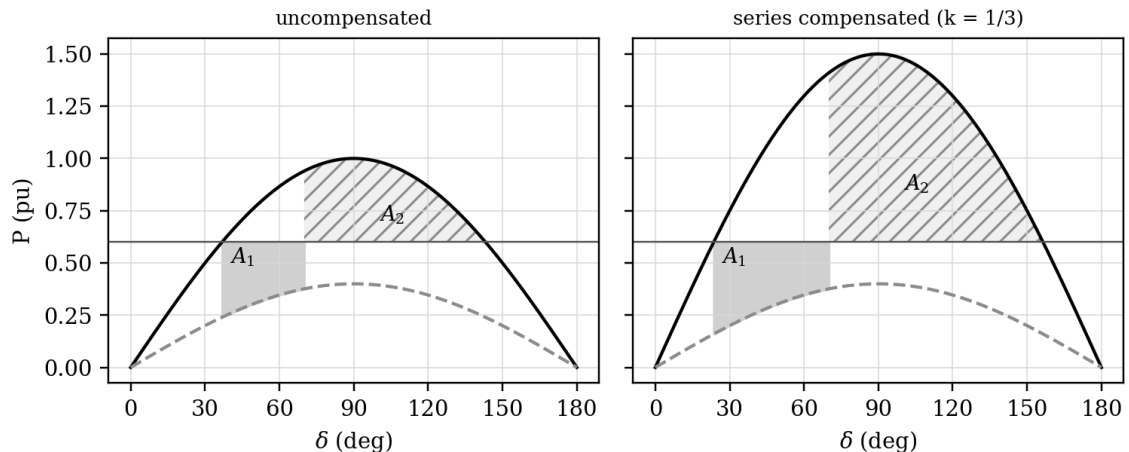


Figure 7.5. Equal-area criterion for an uncompensated line (left) and a series-compensated line with $k = 1/3$ (right).

Power oscillation damping uses the same lever continuously. Transmitted power should be increased while the sending-end machine accelerates (angle increasing) and decreased while it decelerates. Modulating the series reactance in phase with the speed deviation produces a damping torque, and because power varies directly with $1/X_{eff}$, a series device achieves this with a small modulation range. Section 7.4.8 shows a simulated example.

7.1.5 Subsynchronous resonance

A series capacitor forms a resonant circuit with the total inductance of the network it is connected to, which includes generators, transformers and lines. If that total reactance at system frequency f_0 is X , the electrical natural frequency is

$$f_e = \frac{1}{2\pi\sqrt{LC}} = f_0 \sqrt{\frac{X_C}{X}} = f_0 \sqrt{k} \quad (7.5)$$

where f_e is the electrical resonant frequency, L and C the total series inductance and capacitance, and f_0 the system frequency. For practical compensation levels (25% to 75%), f_e lies below f_0 , which is why the phenomenon is called subsynchronous. A subsynchronous current in the stator produces an air-gap field rotating at f_e , which induces rotor currents and torque at the complementary frequency $f_0 - f_e$ (Figure 7.6). If $f_0 - f_e$ coincides with a torsional natural frequency of the turbine-generator shaft, electrical and mechanical oscillations reinforce each other. The torsional mode becomes negatively damped and the shaft can fail. Two shaft failures at the Mohave plant in Nevada, in 1970 and 1971, first demonstrated this mechanism. An induction-generator effect, in which the rotor presents negative resistance at subsynchronous frequencies, can also cause self-excitation, and transient torque amplification can occur after faults even without steady instability.

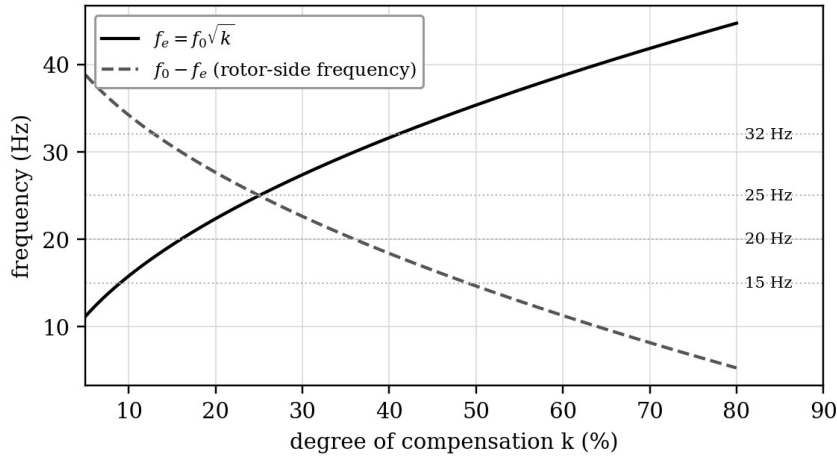


Figure 7.6. Electrical resonant frequency and the complementary rotor frequency versus degree of compensation for a 50 Hz system, with typical torsional mode frequencies.

SSR is the main reason why the degree of fixed compensation near thermal plants is often limited to 30% to 50%, despite the strong transfer benefits of higher values. The thyristor-controlled devices of this chapter change the picture. A controlled TCSC can be designed to modify its subsynchronous impedance, while an SSSC does not rely on a physical series capacitor to create its commanded fundamental-frequency voltage. Neither fact should be interpreted as automatic immunity from all subsynchronous interactions. SSR and SSCI remain system and control dependent. Examples 7.17 and 7.30 compute resonance frequencies.

7.1.6 Practical limits and protection of series capacitors

The capacitor carries full line current and sees the full fault current, so its voltage during a fault can reach several times its rated value. Figure 7.7 shows the standard protective arrangement of a series-capacitor segment. A zinc-oxide varistor (MOV) in parallel clamps the capacitor voltage at about 2 to 2.5 times rated peak and absorbs energy during the fault. A triggered spark gap, with a damping reactor in series, protects the MOV when its energy rating would otherwise be exceeded. A bypass breaker closes for prolonged faults or maintenance. The whole assembly sits on an insulated platform at line potential, with control signals and auxiliary power brought by fibre optics. Its control, measurement and cooling equipment is the same as that used for TCSCs.

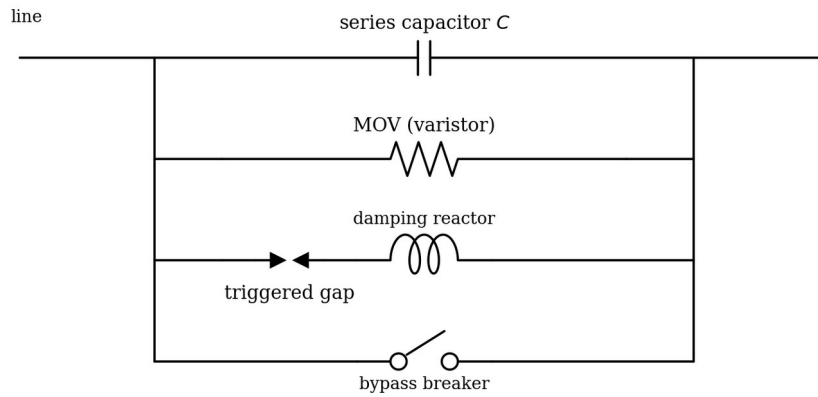


Figure 7.7. Protection of a series-capacitor segment: MOV, triggered gap with damping reactor, and bypass breaker.

The degree of compensation is limited in practice to about 70% to 75%. Higher values make the line current and the capacitor voltage very sensitive to small changes, raise fault levels, distort the voltage profile along the line (Example 7.16 shows the overvoltage at the capacitor terminal) and complicate distance protection, which can be fooled by the negative reactance it sees.

7.1.7 Development of controlled series compensation: a short literature survey

Fixed series capacitors have been in service since the 1950s. Controlled series compensation emerged with the FACTS initiative of the late 1980s. Vithayathil and co-workers conceived the rapid adjustment of network impedance in the mid-1980s, a scheme in which a thyristor-controlled reactor in parallel with a series capacitor varies the line reactance, and reported system case studies at CIGRE in 1988 [1]. The first TCSC installations followed in the United States: Kayenta (WAPA, 1992) inserted a controlled section into a 230 kV fixed bank, and Slatt (BPA, 1993) demonstrated a multi-module 500 kV TCSC. Later projects targeted specific problems: SSR mitigation at Stöde in Sweden, inter-area damping on the Brazilian North-South interconnection, and grid interconnection on the Kanpur-Ballabgarh and Rourkela-Raipur 400 kV lines in India. Hingorani's NGH damper [2] supplied the idea that a thyristor-controlled series capacitor could also be made subsynchronously neutral. Karady and co-workers proposed the GTO-controlled series capacitor in the early 1990s [3], and Gyugyi proposed the SSSC in 1989 as part of the voltage-source family that also includes the STATCOM and UPFC; its full analysis appeared later [4]. The analytical treatments of Hingorani and Gyugyi [5], Mathur and Varma [6], Padiyar [7] and Zhang *et al.* [8], together with the TCSC impedance analyses by Christl *et al.* and Helbing and Karady [9], form the basis of this chapter.

7.2 The GTO-Thyristor Controlled Series Capacitor (GCSC)

7.2.1 Circuit and principle

The GCSC (Figure 7.8) is a capacitor shunted by a bidirectional valve of turn-off devices, GTO thyristors in the original proposal and IGCTs or IGBTs in modern designs. When the valve conducts, the capacitor is short circuited and its voltage is zero. When the valve is open, line current flows into the capacitor and builds up its voltage. The object of control is the fundamental component of the capacitor voltage at a given line current, in other words an adjustable series capacitive reactance.

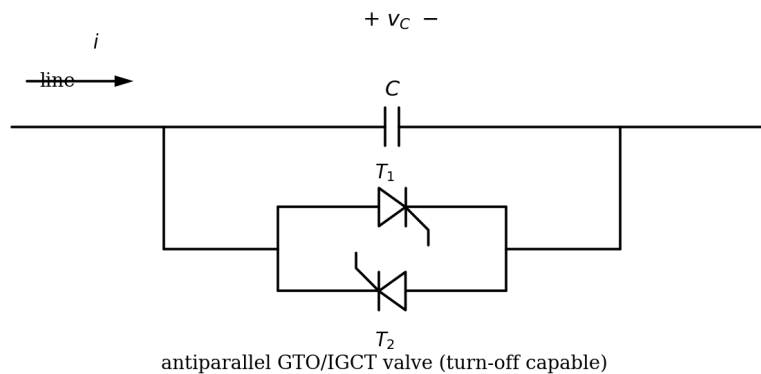


Figure 7.8. GCSC module: series capacitor with an antiparallel turn-off valve.

Let the line current be $i = I \cos \omega t$, so that its crests occur at $\omega t = 0, \pi, 2\pi, \dots$. The valve is opened (turned off) at a delay angle γ after a current crest. From that instant the capacitor integrates the current. The voltage rises to a maximum at the next current zero and returns to zero at $\omega t = \pi - \gamma$, symmetrically placed about the current zero. At that instant the valve is forward biased with zero voltage across it, and it turns on again to keep the capacitor bypassed until γ after the next crest. The cycle repeats with opposite polarity. Figure 7.9 shows the result for three delay angles.

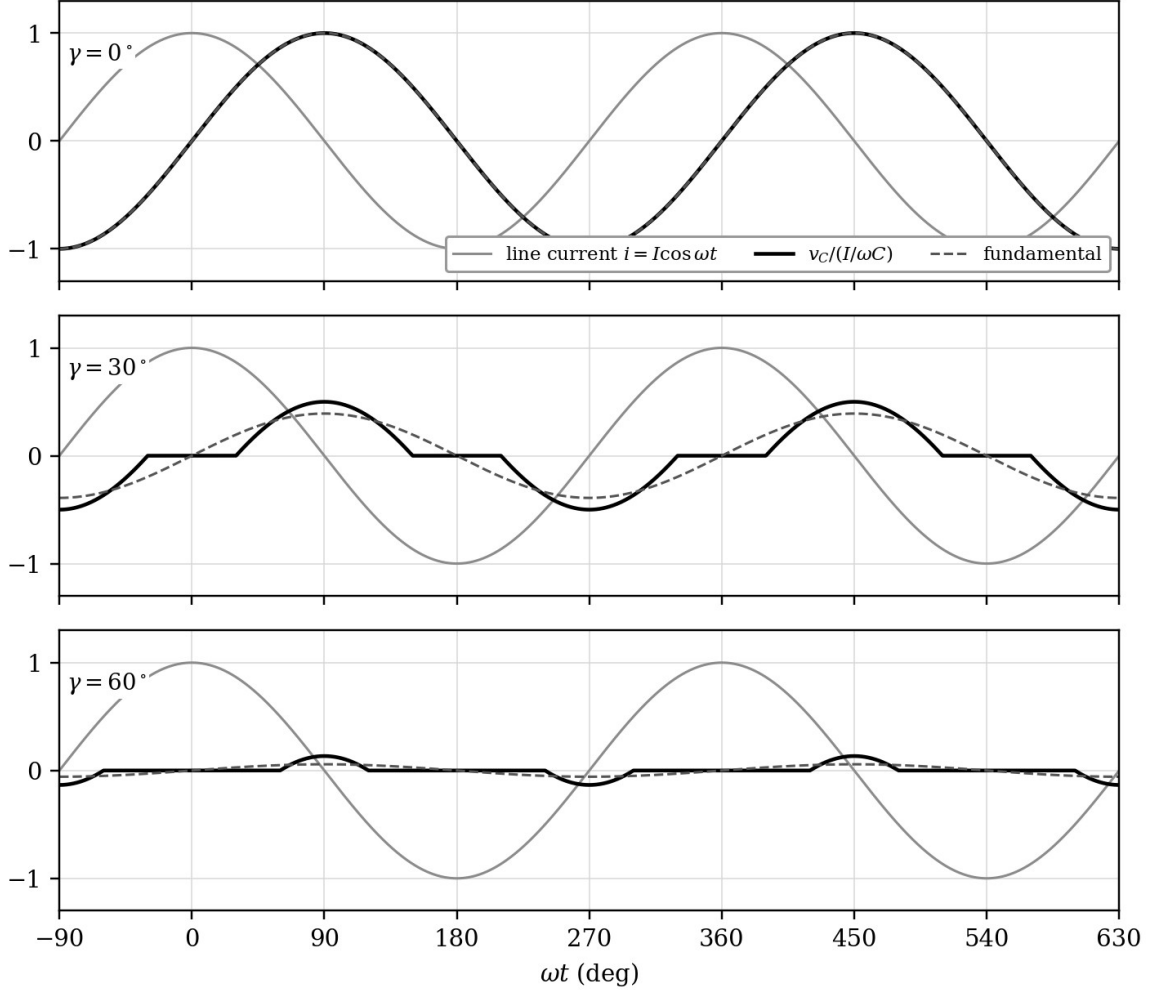


Figure 7.9. GCSC capacitor voltage (with its fundamental) and line current for turn-off delays of 0° , 30° and 60° .

7.2.2 Capacitor voltage

During the insertion interval of the positive half-cycle,

$$v_C(t) = \frac{1}{C} \int_{\gamma/\omega}^t I \cos \omega \tau d\tau = \frac{I}{\omega C} (\sin \omega t - \sin \gamma), \gamma \leq \omega t \leq \pi - \gamma \quad (7.6)$$

where v_C is the capacitor voltage, I the peak line current, C the capacitance and γ the turn-off delay measured from the current crest. Outside this interval $v_C = 0$, and the negative half-cycle is the mirror image. The term $(I/\omega C) \sin \gamma$ is a γ -dependent offset that shifts the sinusoid obtained at $\gamma = 0$ downward, so that the voltage starts and ends at zero. The peak capacitor voltage, at the current zero, is $(I/\omega C)(1 - \sin \gamma)$.

7.2.3 Fundamental reactance and the duality with the TCR

The GCSC is the exact dual of the TCR of Chapter 6. The TCR is a reactor fed from a voltage source, whose valve closes at a voltage-zero-referenced delay and opens naturally at current zero. The GCSC is a capacitor fed from a current source, whose valve opens at a current-crest-referenced delay and closes naturally at voltage zero. Current and voltage, inductance and capacitance, and the closing and opening events all swap roles. The Fourier analysis of the TCR therefore carries over with these substitutions. Integrating Equation (7.6) against the fundamental gives

$$V_{CF}(\gamma) = \frac{I}{\omega C} \left(1 - \frac{2\gamma}{\pi} - \frac{\sin 2\gamma}{\pi} \right) \quad (7.7)$$

where V_{CF} is the peak fundamental capacitor voltage. Its ratio to the current defines the effective capacitive reactance

$$X_C(\gamma) = \frac{V_{CF}}{I} = X_C \left(1 - \frac{2\gamma}{\pi} - \frac{\sin 2\gamma}{\pi} \right), X_C = \frac{1}{\omega C} \quad (7.8)$$

where $X_C(\gamma)$ is the fundamental series reactance and X_C the reactance of the bare capacitor. The capacitor is fully inserted at $\gamma=0$ and fully bypassed at $\gamma=\pi/2$. The angle during which the capacitor is in circuit in each half-cycle, the *blocking angle*, is $\xi=\pi-2\gamma$. Figure 7.10 plots Equation (7.8). As with the TCR, the law is nonlinear and steep near full insertion, so the controller uses an inverse look-up table (Example 7.22).

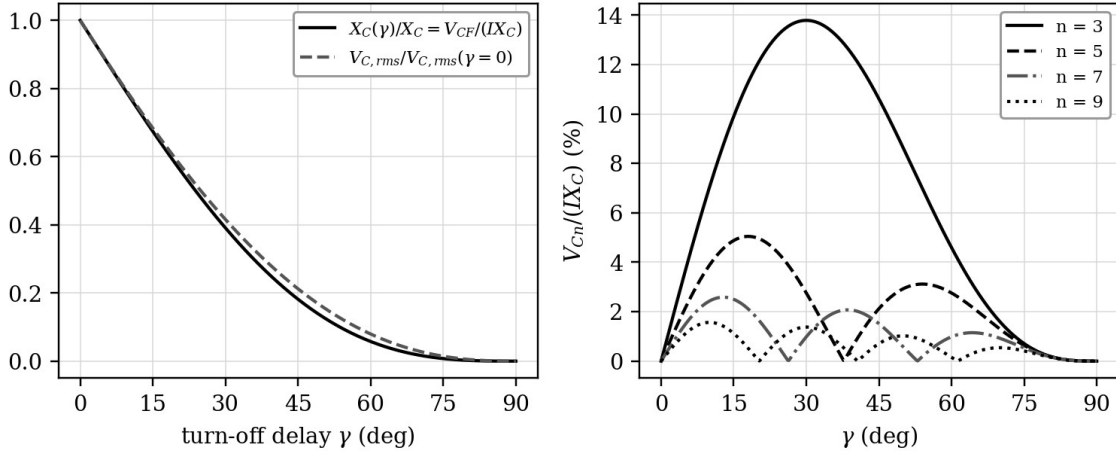


Figure 7.10. GCSC normalized fundamental reactance and RMS voltage versus turn-off delay (left) and harmonic voltages (right).

The total RMS capacitor voltage and the individual harmonics follow by the same duality. With γ playing the role of the TCR's peak-referenced delay,

$$V_{C,rms} = \frac{I}{\omega C} \sqrt{\frac{(\pi-2\gamma) \left(\frac{1}{2} + \sin^2 \gamma \right) - \frac{3}{2} \sin 2\gamma}{\pi}} \quad (7.9)$$

$$V_{Cn} = \frac{I}{\omega C} \cdot \frac{4}{\pi} \cdot \frac{\sin \gamma \cos n\gamma - n \cos \gamma \sin n\gamma}{n(n^2-1)}, n=3, 5, 7, \dots \quad (7.10)$$

where $V_{C,rms}$ is the total RMS capacitor voltage and V_{Cn} the peak n th harmonic voltage (its magnitude is used). The third harmonic reaches its largest value of 13.8% of IX_C at $\gamma=30^\circ$, mirroring the TCR at $\alpha=120^\circ$. The RMS expression contains $\sin^2 \gamma$, not $\sin^2 2\gamma$. The two forms agree only at $\gamma=60^\circ$, where $\sin 2\gamma = \sin \gamma$. Elsewhere the second form is wrong: it gives more than twice the true RMS voltage at $\gamma=30^\circ$, and its radicand becomes negative beyond about 60° (Example 7.21).

7.2.4 Operating modes and characteristics

A GCSC is operated in one of two modes. In the *voltage compensating mode*, the device maintains a rated compensating voltage V_{Cmax} over a range of line currents $I_{min} \leq I \leq I_{max}$. The capacitor is chosen so that the full capacitor produces V_{Cmax} at the minimum current, $X_C = V_{Cmax}/I_{min}$; at higher currents γ is increased to keep the fundamental voltage constant. In the *impedance compensating mode*, the device maintains a chosen reactance at any current up to I_{max} ; the capacitor is then sized for $X_C = V_{Cmax}/I_{max}$, and any reactance between zero and X_C is available. Figure 7.11 shows the two V-I characteristics. Losses are mainly conduction losses of the valve, which carries the line current while the capacitor is bypassed. They are therefore largest at small reactance and high current, which is the opposite of the TCSC (Section 7.4.7).

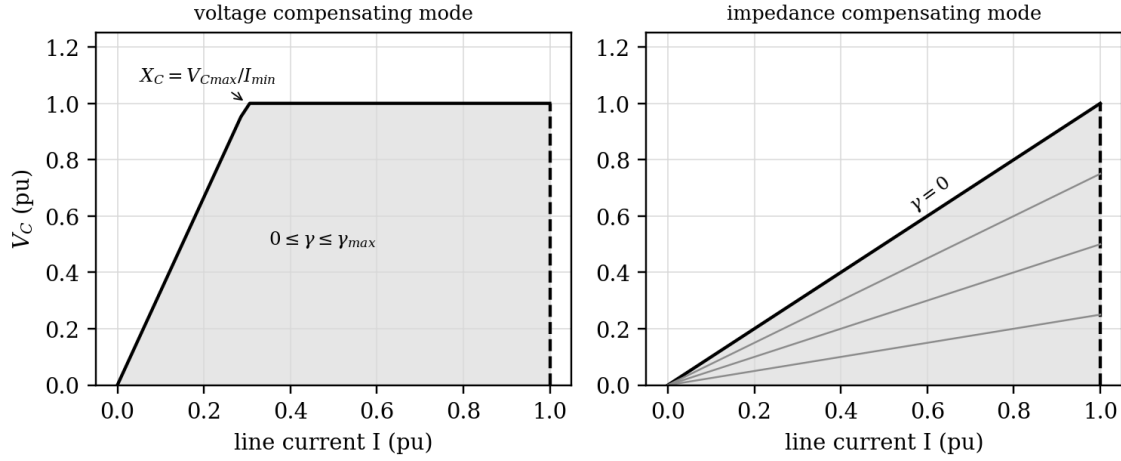


Figure 7.11. GCSC V-I operating areas in voltage compensating mode (left) and impedance compensating mode (right).

Example 7.2. Effective reactance of a GCSC at a given turn-off delay

A GCSC has a fully inserted capacitive reactance of $8 \, \Omega$. Find the effective fundamental reactance at $\gamma = 30^\circ$, measured from the current crest.

Solution. From Equation (7.8) with $\gamma = \pi/6$, the bracket is $1 - 1/3 - \sin(\pi/3)/\pi = 0.3910$, so $X_c(\gamma) = 8 \times 0.3910 = 3.13 \, \Omega$.

Note. The angle reference is part of the equation. A delay measured from a current zero must be converted ($\gamma = \gamma_0 - 90^\circ$) before Equation (7.8) is used.

7.2.5 Harmonics, sequential control and practical aspects

Because a GCSC is inserted directly in the line without a transformer, the multi-pulse cancellation used in converters is not readily available. A small number of publications propose six- and twelve-pulse arrangements through series transformers, but the standard remedy is *sequential control*. The total capacitor is divided into m series modules, each rated $1/m$ of the total voltage. All but one module are either fully inserted or fully bypassed, and only one module is delay-angle controlled. The harmonic voltages, expressed on the total rating, then fall by roughly the factor m , exactly as for the segmented TCR (Example 7.25). Splitting the high-voltage valve into four or more modules is also attractive on its own, since each valve then sees a fraction of the total voltage.

The GCSC is principally encountered in the literature as a conceptual and prototype-level controlled-series-capacitor topology; transmission practice has been dominated by fixed series capacitors, TCSC and, increasingly, converter-based series compensation. Turn-off semiconductors have lower surge capability than thyristors, and the valve must interrupt full line current twice per cycle. A short-circuit or fault current through the valve requires a rapid thyristor or mechanical bypass. The concept remains valuable both for its clean theory, as the dual of the TCR, and in distribution-level series compensators where IGBTs are adequate.

7.2.6 Gate timing, valve stress and practical control

The GCSC control variable is not a conventional thyristor firing delay. The bidirectional turn-off valve conducts while the capacitor is bypassed and is commanded off after the selected current crest. From that instant until the valve turns on again at zero capacitor voltage, the line current charges the capacitor. A practical controller therefore begins with current synchronization, calculates the requested fundamental reactance, converts it into a turn-off delay through the inverse of Equation (7.8), and issues turn-off commands under protection supervision. The current reference must be filtered enough to reject harmonics without introducing a phase error large enough to shift the intended insertion interval.

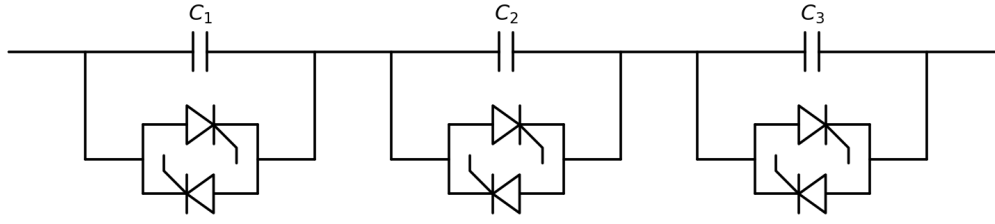
The valve has two complementary duties. During bypass it carries the line current and produces conduction loss. During the open interval it must withstand the capacitor voltage. The blocking-voltage rating is therefore linked to maximum inserted capacitor voltage, whereas current rating is linked to corridor current and fault-bypass duty. Modularization reduces the blocking voltage per valve and enables sequential control in which most modules are either fully inserted or fully bypassed and only one module is partially controlled.

The waveform should be interpreted physically. At full insertion the device behaves as an ordinary series capacitor. As the bypass interval is enlarged, capacitor voltage is clamped to zero for a larger fraction of each half-cycle. The fundamental component decreases continuously, but odd-harmonic voltage changes nonlinearly with delay. Fundamental-reactance control and harmonic-voltage duty must therefore be checked together.

7.3 The Thyristor-Switched Series Capacitor (TSSC)

7.3.1 Circuit and principle

A TSSC (Figure 7.12) consists of several capacitors in series, each shunted by a bypass valve made of antiparallel thyristor strings. The structure resembles the sequentially controlled GCSC, but the valves are ordinary thyristors, which cannot be turned off by the gate. A capacitor is *inserted* by withdrawing the gate pulses of its valve, and the valve then stops conducting at the next current zero. It is *bypassed* by firing the valve. The degree of compensation therefore changes in steps, set by the number of stages in service, and each stage is either fully in or fully out. No partial conduction occurs, so a TSSC generates no harmonics in steady state.



each stage: capacitor + antiparallel thyristor bypass valve

Figure 7.12. TSSC with three stages, each a capacitor with an antiparallel thyristor bypass valve.

7.3.2 Insertion at current zero and the DC offset

Since a thyristor can only commutate naturally, a stage is necessarily inserted at a zero crossing of the line current. Taking $i = I_m \sin \omega t$ with a discharged capacitor inserted at $\omega t = 0$,

$$v_C(t) = \frac{1}{C} \int_0^t I_m \sin \omega \tau d\tau = I_m X_C (1 - \cos \omega t) \quad (7.11)$$

where I_m is the peak line current and X_C the stage reactance. The capacitor voltage therefore contains a DC offset equal to the amplitude of its AC component, and it swings between zero and $2I_m X_C$ (Figure 7.13a). In an ideal lossless circuit the offset would persist; in practice it decays slowly through the capacitor's leakage and the network's resistance. Conversely, a stage can be inserted without offset only if the capacitor already holds the voltage $-I_m X_C$, the value its steady-state waveform would have at that current zero (Figure 7.13b).

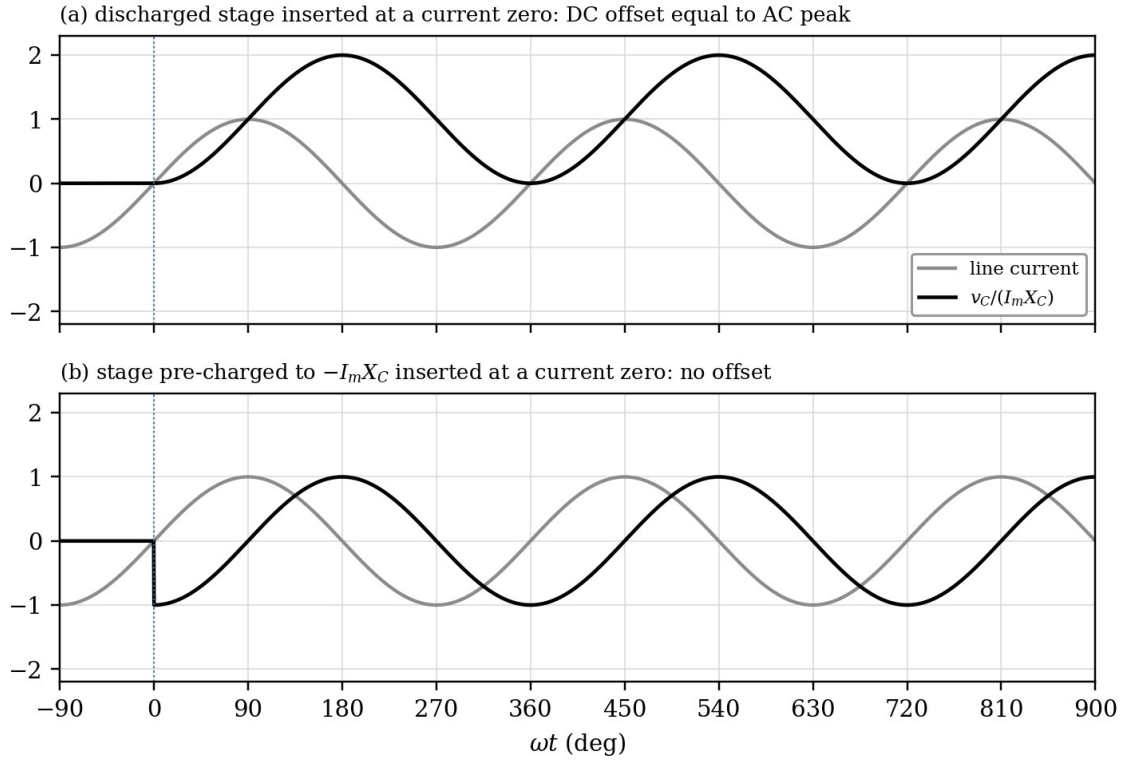


Figure 7.13. TSSC stage insertion at a current zero: (a) discharged capacitor, with DC offset; (b) capacitor pre-charged to $-I_m X_C$, without offset.

The bypass rule is the counterpart for the valve. Firing the valve when the capacitor holds a voltage V_{C0} discharges the capacitor through the small inductance L_v of the valve circuit, giving a surge current of peak value

$$I_{surge} \approx \frac{V_{C0}}{Z_0}, Z_0 = \sqrt{\frac{L_v}{C}} \quad (7.12)$$

where Z_0 is the characteristic impedance of the discharge loop and L_v its inductance. To keep the valve stress low, the valve is fired when the capacitor voltage passes through zero. With a DC offset present, this happens only once per cycle, at the current zero where the voltage returns to zero. In the steady state without offset it happens at every crest of the line current. A current-limiting reactor in the valve branch limits di/dt to the thyristor's capability (Example 7.28).

7.3.3 Operating modes and characteristics

In the *voltage compensating mode*, the TSSC is asked to approximate a rated compensating voltage over a range of line current. If m equal stages of reactance X_C are provided, the rated voltage at the minimum current is obtained with all stages in service, $V_{Cmax} = m X_C I_{min}$; as current rises, stages are bypassed to keep the voltage at or below V_{Cmax} . The frequently quoted relation $V_{Cmax} = 4 X_C I_{min}$ is simply the case $m=4$. In the *impedance compensating mode*, the number of stages in service is held constant, giving a fixed series reactance at any current up to I_{max} . Figure 7.14 shows the two stepped characteristics, which are the discrete counterparts of Figure 7.11.

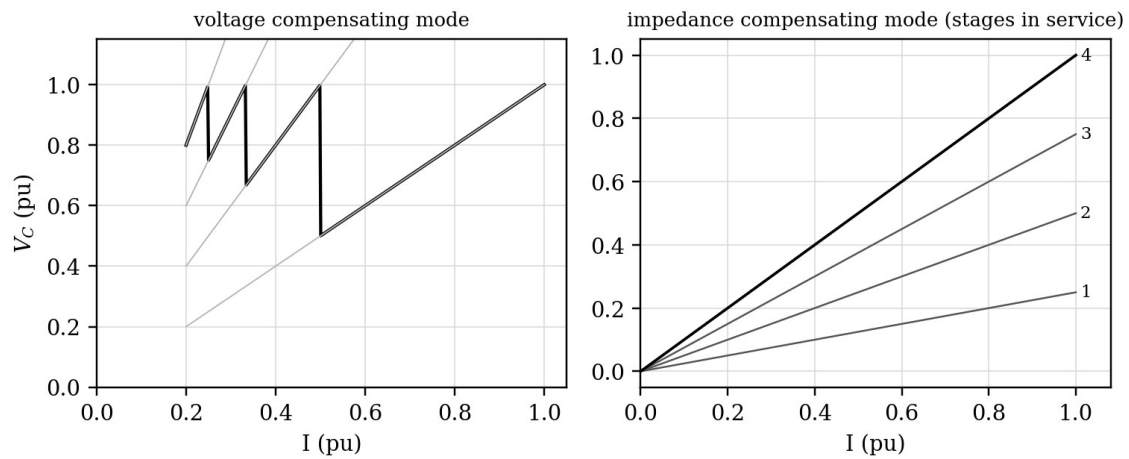


Figure 7.14. TSSC V-I characteristics: voltage compensating mode with stage switching (left) and impedance compensating mode for 1 to 4 stages (right).

Losses arise only in the valves of bypassed stages, which carry the full line current. They are therefore largest with few stages inserted and small with all stages in service. Each valve must block the peak capacitor voltage, including any offset, and carry the full line current continuously and the fault current for the duration set by the protection.

Example 7.3. Distinct compensation levels of an equal-stage TSSC

A TSSC has four equal stages of $3\ \Omega$ each and is installed in a line of $30\ \Omega$. Find the available degrees of compensation.

Solution. The inserted reactance can be 0, 3, 6, 9 or $12\ \Omega$, giving 0%, 10%, 20%, 30% and 40%. Although four bypass valves allow $2^4 = 16$ switching combinations, equal stages produce only five distinct levels.

Note. Binary-weighted stages of 1.5, 3, 6 and $12\ \Omega$ would give sixteen levels in 5% steps from 0% to 75% with the same number of valves; their capacitances are weighted in the inverse ratio.

7.3.4 Response and SSR

A TSSC responds within half a cycle to a bypass command and within one cycle to an insertion command, and because its stages can be switched alternately it is adequate for power-flow scheduling and, with suitable control, for power oscillation damping. Its weakness is subsynchronous resonance. An inserted TSSC stage is simply a capacitor and behaves exactly like a fixed series capacitor at subsynchronous frequencies. In principle, switching could be modulated to counteract subsynchronous oscillations, but the step-wise, cycle-delayed action is too coarse for reliable SSR damping. Pure TSSC schemes are therefore not used where a high degree of compensation is required near turbine-generators with vulnerable torsional modes. In such cases a TCSC, or a TSSC combined with a TCSC module, is preferred. Examples 7.26 to 7.30 cover TSSC sizing, switching and SSR screening.

7.3.5 Stage coordination, binary weighting and switching logic

Equal TSSC stages are easy to manufacture and share voltage uniformly, but n equal stages provide only $n+1$ distinct reactance levels. Binary-weighted reactances X , $2X$, $4X$ and so on provide 2^n levels with the same number of valves. Because capacitor reactance is inversely proportional to capacitance, the capacitances are weighted in the opposite order, C , $C/2$, $C/4$ and so on. This is the reverse of a binary shunt capacitor bank, in which the capacitance (susceptance) itself is weighted.

A supervisory controller chooses the combination of inserted stages from a power-flow, current, voltage or impedance reference. The requested change cannot be imposed at an arbitrary instant. Insertion waits for natural current extinction of the bypass valve, while low-stress bypass is coordinated with capacitor voltage. When several stages change state, their operations can be staggered to reduce simultaneous voltage steps and platform stress. The response is therefore discrete in magnitude and constrained by natural commutation in time.

7.3.6 Protection, ratings and practical limitations

Each inserted capacitor carries full line current and must be rated for steady-state voltage, insertion offset and temporary overload. Each bypass valve carries full line current while its stage is out of service and must withstand

the capacitor voltage while the stage is inserted. A small reactor in the valve branch limits the discharge current and di/dt when a charged stage is bypassed (Equation (7.12)). MOV protection and a mechanical bypass path are coordinated with the series-capacitor protection described in Section 7.1.6.

The attraction of the TSSC is its simplicity. Once a stage is fully inserted, it is a linear capacitor and does not create phase-control harmonics. Its limitations are equally important: compensation is quantized, switching is constrained by natural commutation, and an inserted stage retains the frequency-dependent behaviour of a physical series capacitor. TSSC is therefore appropriate for stepped scheduling and coarse power-flow redistribution, while TCSC or SSSC is preferable when continuously variable damping or wide-range modulation is required.

7.4 The Thyristor-Controlled Series Capacitor (TCSC)

7.4.1 Circuit, purpose and installations

The TCSC (Figure 7.15) places a thyristor-controlled reactor in parallel with a series capacitor. The combination is a tunable parallel LC circuit carrying the line current. By varying the conduction of the reactor, the controller can raise the effective capacitive reactance above that of the capacitor alone (the *capacitive vernier* or boost range), leave the capacitor alone (*blocking*), short it through the reactor (*bypass*), or make the module net inductive (*inductive vernier*). In a typical installation the TCSC module forms only part of the total compensation: a larger fixed series capacitor provides the bulk of the reactance cheaply, and the TCSC adds controllability. The module shares the platform-mounted MOV, triggered gap and bypass breaker of Figure 7.7, but because the thyristor valve can bypass the capacitor within a fraction of a cycle, the gap is often unnecessary.

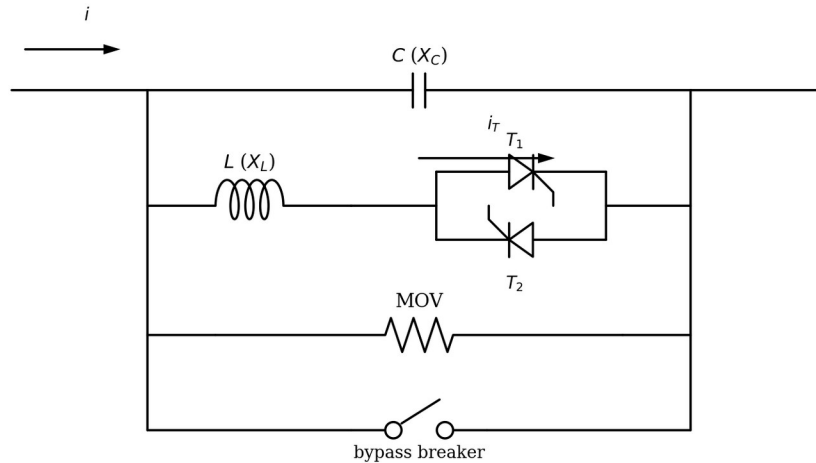


Figure 7.15. TCSC module: capacitor in parallel with a reactor and antiparallel thyristor valve, with MOV and bypass breaker.

Table 7.2 summarizes representative TCSC installations reported in the FACTS literature. The table is intended to show the range of early applications, including power-flow control, SSR mitigation and oscillation damping, rather than to serve as an equipment procurement specification.

Table 7.2 Representative TCSC installations.

Installation	Year	Voltage	Purpose
Kayenta, WAPA, USA	1992	230 kV	First TCSC; controlled section within a fixed bank, power-flow control
Slatt, BPA, USA	1993	500 kV	Six-module prototype; SSR and control studies
Stöde, Sweden	1998	400 kV	SSR mitigation for nearby generation
North-South interconnection, Brazil	1999	500 kV	Damping of inter-area oscillations
Kanpur-Ballabgarh, India	2000s, in two phases	400 kV	First indigenous TCSC (BHEL with POWERGRID); a thyristor-controlled section varies an 8% fixed segment up to 20%
Rourkela-Raipur, India	2004	400 kV, double circuit	First TCSC in India; East-West inter-regional transfer and damping

7.4.2 The simplified (voltage-source) model

The first, and still the most widely quoted, analysis treats the TCR as if it were connected to a sinusoidal voltage source, exactly as in Chapter 6. Let α be the firing delay measured from the crest of the capacitor voltage, so that each thyristor conducts for $2\beta = \pi - 2\alpha$, where $\beta = \pi/2 - \alpha$ is the half conduction angle. The TCR then behaves as an inductive reactance

$$X_L(\alpha) = X_L \frac{\pi}{\pi - 2\alpha - \sin 2\alpha} = X_L \frac{\pi}{2\beta - \sin 2\beta}, \quad X_L = \omega L \quad (7.13)$$

where $X_L(\alpha)$ is the equivalent reactance of the controlled reactor and X_L its full-conduction value. In parallel with the capacitor, and using the signed convention in which positive reactance is inductive,

$$X_{TCSC}(\alpha) = \frac{X_C X_L(\alpha)}{X_C - X_L(\alpha)} \quad (7.14)$$

where X_{TCSC} is the signed reactance of the module. With $X_L < X_C$, the module is inductive for small α , passes through a parallel resonance where $X_L(\alpha) = X_C$, and becomes capacitive with a reactance magnitude larger than X_C beyond it, falling back to $-X_C$ as $\alpha \rightarrow 90^\circ$. The ratio of reactor to line current follows from current division, $I_{TCR}/I = X_C/(X_C - X_L(\alpha))$. Close to resonance this ratio is large, which is why the capacitive boost must be limited (Example 7.32).

7.4.3 Why the simplified model is not accurate

The simplified model assumes that the reactor sees a sinusoidal voltage. In a TCSC it does not. The module is driven by the *line current*, which the rest of the network imposes almost independently of what the TCSC does. The capacitor voltage is shaped by the reactor pulses themselves. When a thyristor fires, the charged capacitor discharges through the reactor in a resonant half-period of the LC loop, reversing its charge, while the line current continues to flow. This *charge reversal* is the central mechanism of the TCSC, and it is absent from the voltage-source picture. The consequences are not small. For the module of Example 7.31 ($C = 1000 \mu\text{F}$, $L = 1.5 \text{ mH}$), the simplified model places the resonance at $\alpha = 48.2^\circ$, whereas the exact analysis places it at 55.4° . At $\alpha = 50^\circ$ the simplified model predicts a capacitive reactance of 27.6Ω , while the actual module is *inductive*, $+5.13 \Omega$. A design based on the simplified model can therefore select an operating point on the wrong side of resonance.

Example 7.4. Locating the simplified TCSC parallel-resonance point

A TCSC has $X_C = 20 \Omega$ and a reactor of $X_L = 4 \Omega$. Use the simplified model, Equation (7.13), to find the firing angle at which parallel resonance occurs, and compare it with the exact current-driven model.

Solution. In the simplified model resonance occurs when $X_L(\alpha) = X_C$, that is when $\pi - 2\alpha - \sin 2\alpha = \pi X_L/X_C = \pi/5$. Solving numerically gives $\alpha = 43.40^\circ$ ($\beta = 46.60^\circ$). The exact model has its first resonance at $\beta_r = \pi/(2\varpi)$ with $\varpi = \sqrt{X_C/X_L} = 2.236$, so $\beta_r = 40.25^\circ$ and $\alpha_r = 49.75^\circ$. The two models place the resonance 6.4° apart.

Note. Between $\alpha = 43.4^\circ$ and 49.8° the two models disagree on whether the module is inductive or capacitive, so the permitted firing band and its resonance margin must be set from the exact characteristic.

7.4.4 Exact analysis of the line-current-driven TCSC

Let the line current be $i = I_m \sin \theta$, $\theta = \omega t$, and let v_C be the capacitor voltage and i_T the thyristor current. While both thyristors block, $C dv_C/dt = i$. While a thyristor conducts,

$$C \frac{dv_C}{dt} = i - i_T, \quad L \frac{di_T}{dt} = v_C \Rightarrow \frac{d^2 v_C}{dt^2} + \omega_0^2 v_C = \frac{1}{C} \frac{di}{dt}, \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad (7.15)$$

where ω_0 is the natural frequency of the LC loop. Its solution during conduction is the sum of a forced term at line frequency and a free oscillation at ω_0 ; the thyristor current ends naturally when it reaches zero. In periodic steady state the conduction interval of each thyristor is centred on the zero crossing of the capacitor voltage, with half-

width β . Matching the capacitor voltage and reactor current at the firing and extinction instants and extracting the fundamental of v_C gives the closed-form reactance [9]:

$$X_{TCSC}(\beta) = -X_C + C_1(2\beta + \sin 2\beta) - C_2 \cos^2 \beta (\varpi \tan \varpi \beta - \tan \beta) \quad (7.16)$$

with

$$\varpi = \frac{\omega_0}{\omega} = \sqrt{\frac{X_C}{X_L}}, X_{LC} = \frac{X_C X_L}{X_C - X_L}, C_1 = \frac{X_C + X_{LC}}{\pi}, C_2 = \frac{4 X_{LC}^2}{\pi X_L} \quad (7.17)$$

where ϖ is the tuning ratio of the module, X_{LC} the parallel reactance of capacitor and full reactor, and C_1, C_2 are constants of the module. Equation (7.16) gives $-X_C$ at $\beta=0$ (blocked) and the full-conduction parallel reactance X_{LC} at $\beta=\pi/2$ (bypass). It becomes infinite where $\tan \varpi \beta$ does, so the resonances lie at

$$\beta_r = \frac{(2k-1)\pi}{2\varpi}, k=1, 2, \dots \quad (7.18)$$

For $1 < \varpi < 3$ ($1/9 < X_L/X_C < 1$) exactly one resonance lies between blocking and full conduction, so the characteristic has one capacitive and one inductive branch. Practical modules use ϖ between about 2 and 3 ($0.11 \leq X_L/X_C \leq 0.25$). For $\varpi > 3$ a second resonance, at $\beta = 3\pi/(2\varpi)$, falls inside the control range close to full conduction (Example 7.33). We verified Equation (7.16) against a direct time-domain solution of Equation (7.15) over the full range of β ; the two agree to better than 0.01%. Figure 7.16 compares the exact and simplified characteristics for $X_L/X_C = 0.133$.

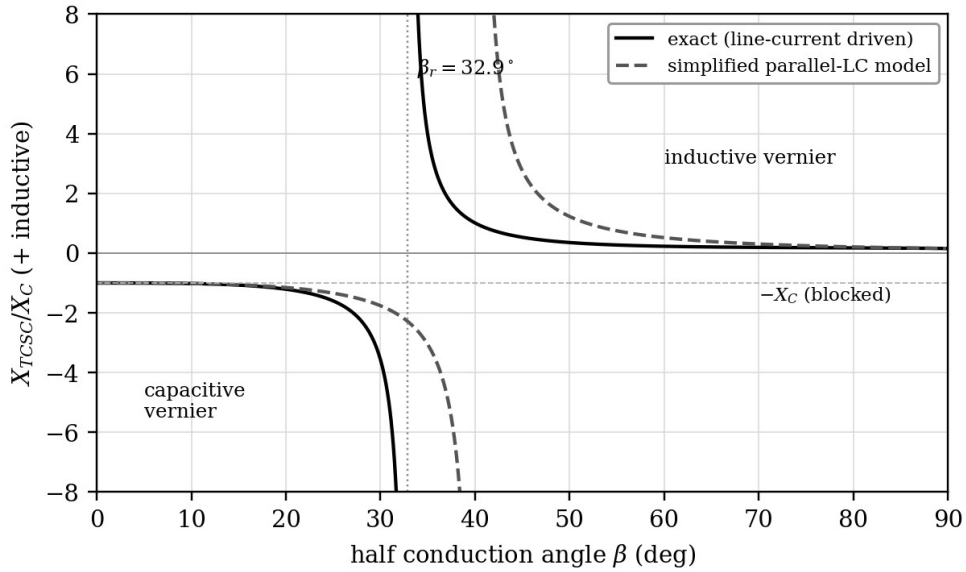


Figure 7.16. TCSC reactance (normalized to X_C , positive inductive) versus half conduction angle: exact line-current-driven characteristic and simplified parallel-LC model.

7.4.5 Waveforms and the charge-reversal mechanism

Figures 7.17 and 7.18 show steady-state waveforms obtained from the exact model for a module with $X_L/X_C = 0.133$. In the capacitive vernier ($\beta = 25^\circ$), each thyristor fires shortly before the capacitor voltage crosses zero. The reactor current pulse reaches about 1.7 times the peak line current and flows against it, so during the reversal the capacitor current is boosted to about 2.7 times the peak line current, and the capacitor voltage is pushed further than the line current alone would push it. The fundamental voltage is 1.6 times that of the bare capacitor. In the inductive vernier ($\beta = 55^\circ$), conduction is long, the reactor carries more than the line current, and the capacitor voltage fundamental leads the current: the module is net inductive at $0.27 X_C$.

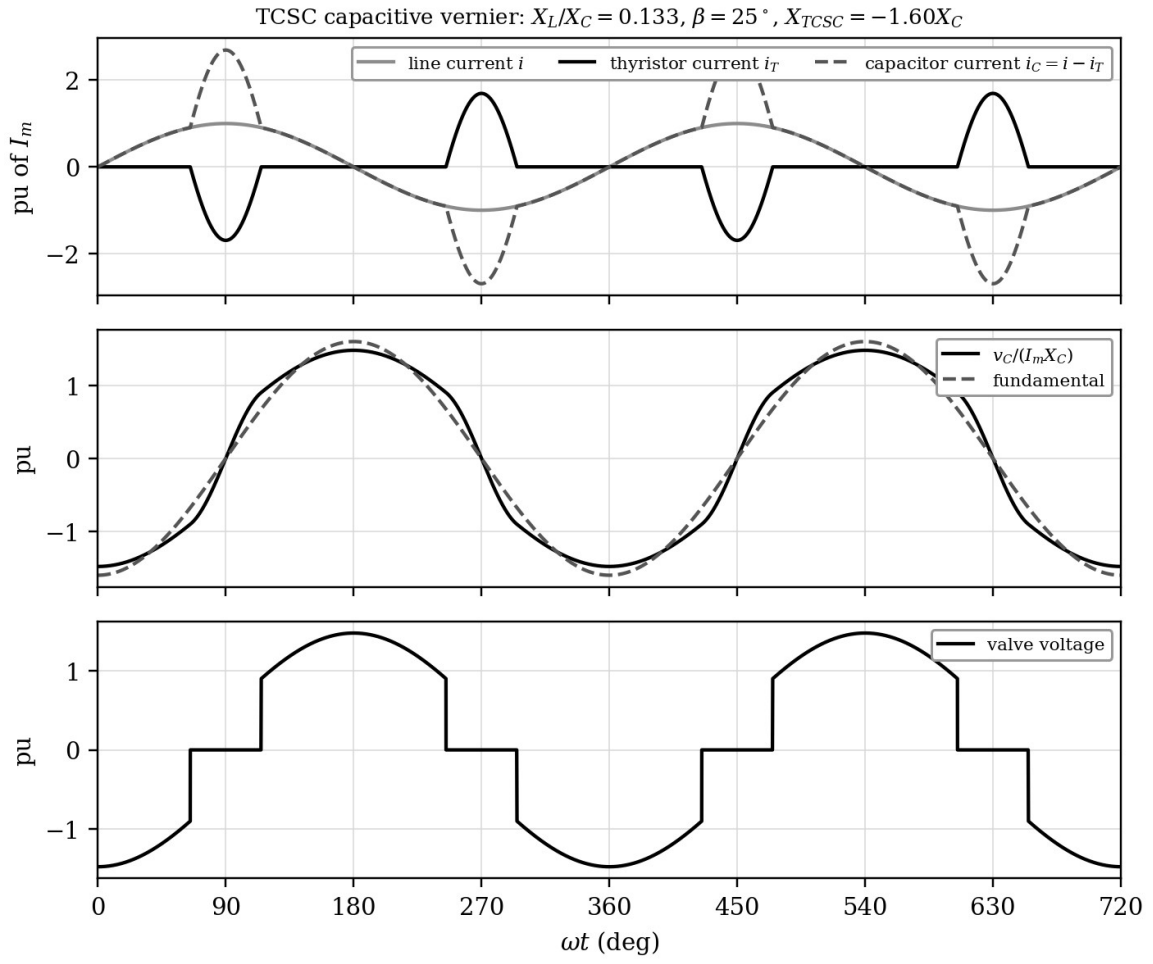
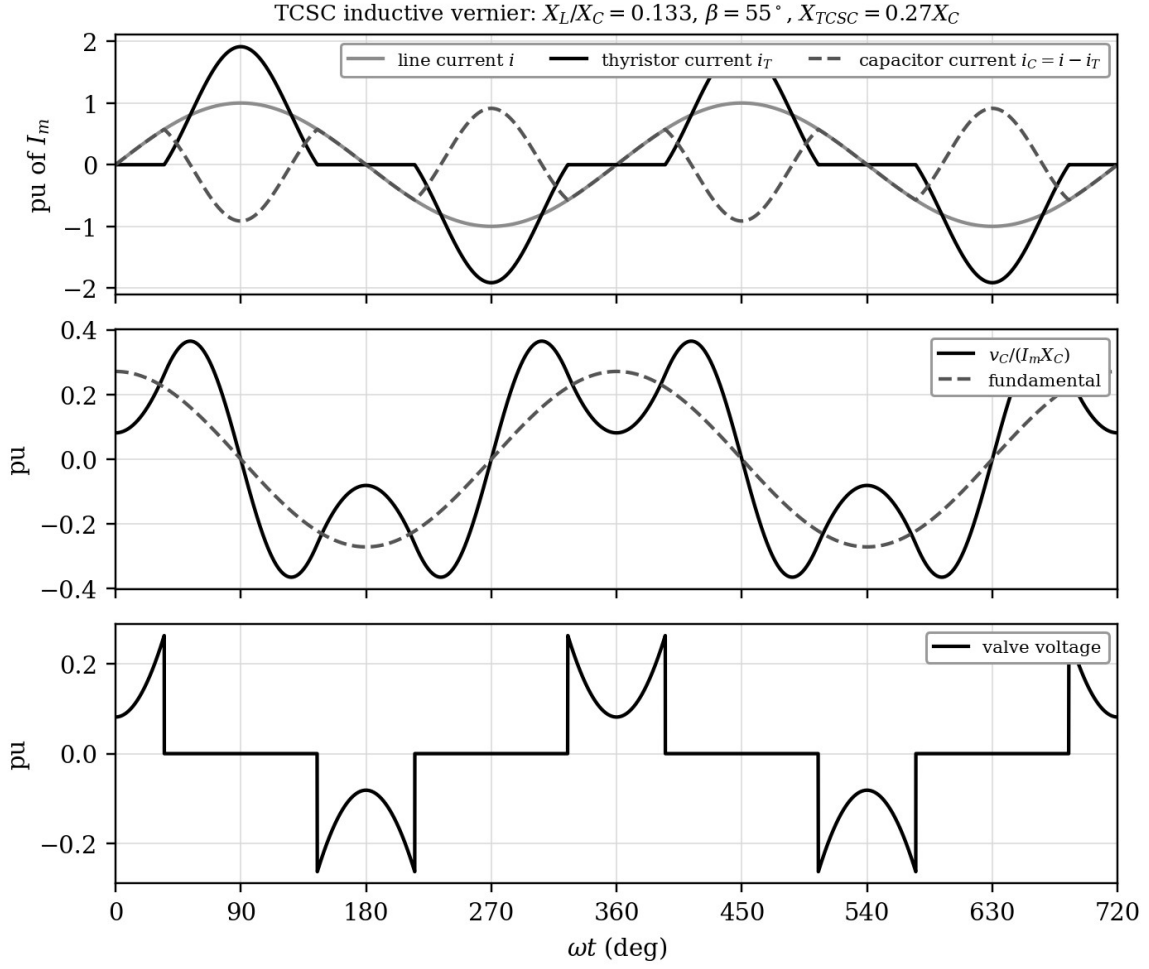


Figure 7.17. TCSC in the capacitive vernier ($\beta = 25^\circ$): line, thyristor and capacitor currents; capacitor voltage with its fundamental; valve voltage.


 Figure 7.18. TCSC in the inductive vernier ($\beta = 55^\circ$).

A helpful idealization, due to Hingorani and Gyugyi [5], is to imagine an infinitesimally small reactor so that the charge reversal is instantaneous. The compensating voltage then splits into two components. One is the *uncontrolled* sinusoid that the line current produces in the capacitor, proportional to the current. The other is a *controlled* square-wave-like component, created by the reversals, whose amplitude depends on the timing of the reversal and is independent of current. Advancing the gating by an angle ξ in successive cycles builds up a DC-like offset that raises the capacitor voltage; retarding it reduces the voltage. With a finite reactor the reversal takes a finite time, and the circuit behaves as if a conventional delay angle controlled it. This picture explains why a TCSC, unlike a simple capacitor, can hold a compensating voltage at low line current, and it links the TCSC to the voltage-source view of series compensation.

7.4.6 Choice of the reactor and harmonics

The ratio X_L/X_C is a central design decision. A small reactor gives a fast, well-defined charge reversal and an effective protective bypass, because the valve then forms a low-impedance path around the capacitor. A larger reactor reduces the peak and RMS thyristor current and the harmonics injected into the line, and widens the usable angle range, but it slows the reversal. Prototype installations used $X_L/X_C \approx 0.13$, and practical values lie between about 0.11 and 0.25 (ω between 2 and 3); below $X_L/X_C = 1/9$ a second resonance appears near full conduction. The resonance angle β_r of Equation (7.18) must sit well inside the control range, with an exclusion band around it.

TCSC harmonics are voltages, not currents: the harmonic currents of the TCR circulate in the LC loop and appear as harmonic voltages across the module. Figure 7.19 shows their magnitude from the exact waveforms. In the normal capacitive vernier range with modest boost, the third harmonic remains a few percent of $I_m X_C$, but it rises steeply as the operating point approaches resonance, which is another reason for limiting the boost.

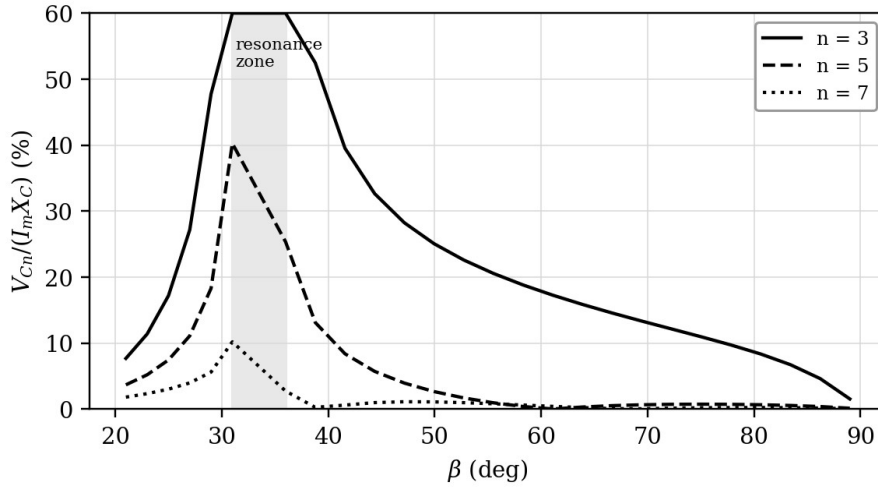


Figure 7.19. Harmonic voltages of the TCSC (exact model, $X_L/X_C=0.133$) as a percentage of $I_m X_C$; the band around resonance is excluded.

7.4.7 Operating modes, characteristics and losses

Four modes are defined. In the *blocking* mode the valve does not conduct and the module is a plain capacitor. In the *bypass* mode the valve conducts continuously and the module presents the small inductive reactance X_{LC} ; this mode is used to reduce stress during faults and to switch the module out. The *capacitive vernier* operates between blocking and a maximum boost, typically $k_B = |X_{TCSC}|/X_C$ of 2 to 3. The *inductive vernier*, between resonance and bypass, is used less often, mainly for fault-current limiting or power-flow reduction.

The attainable range depends on current (Figure 7.20). At low line current the maximum boost is set by the angle limit α_{Clim} . At high current it is set by the capacitor voltage rating, so the attainable reactance falls as V_{Cmax}/I . A range of uncontrollable reactance remains between the bare capacitor (X_{Cmin}) and the smallest inductive value (X_{Lmin}). Modularization, with several smaller TCSCs in series, narrows these gaps. Losses come almost entirely from the thyristor and reactor currents. They are small in blocking and increase with boost, since the reactor current rises sharply towards resonance (Example 7.36).

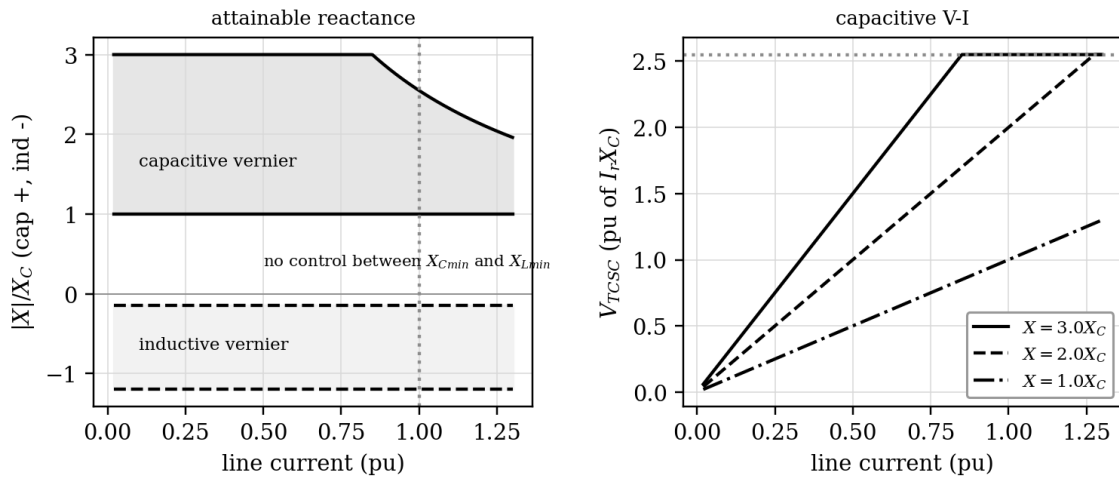


Figure 7.20. TCSC attainable reactance versus line current (left) and capacitive V-I characteristic for constant-reactance operation (right).

7.4.8 Control

Internal control. The internal control converts a reactance reference into firing instants. Two synchronization approaches are used. The first locks a PLL onto the fundamental of the line current, which must be heavily filtered so that subsynchronous and supersynchronous components do not disturb the firing. The second predicts the zero crossing of the capacitor voltage from the measured voltage and current and fires a set angle before it, which keeps the reversals equally spaced at system frequency. That spacing is the property that makes the TCSC

subsynchronously neutral (Section 7.4.9). A reactance estimator, $\hat{X} = V_{C1}/I_1$, closes an outer trim loop around the inverse characteristic.

External control. Figure 7.21 shows the external structure. Depending on the application, the regulated quantity is line power or current (power-flow control), the transmission angle, or a reactance schedule. Supplementary loops add power oscillation damping and, if required, SSR damping, and a transient-stability function drives the module to maximum boost immediately after a fault.

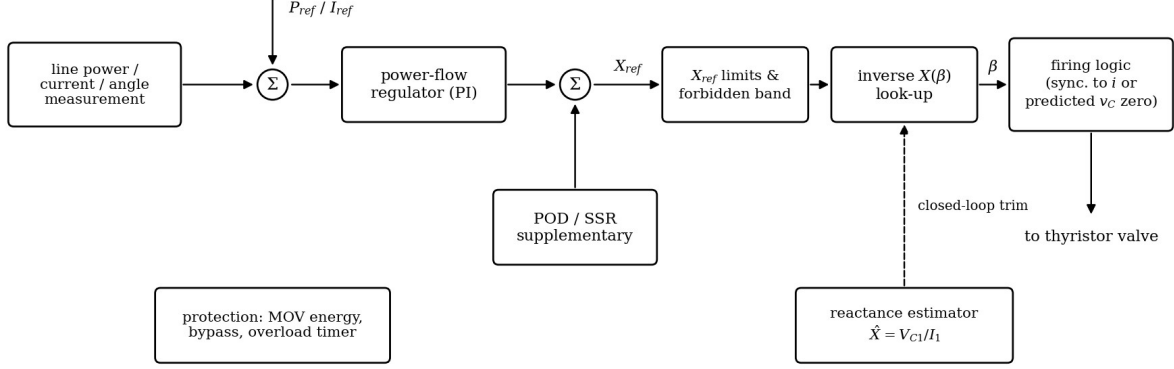


Figure 7.21. TCSC external and internal control structure.

The complete TCSC controller solves two problems, an external one and an internal one. The external controller decides what the corridor requires, such as a line-power order, scheduled reactance, transient-stability boost or damping modulation. The internal controller realizes that request without crossing the resonance exclusion zone or violating capacitor-voltage, reactor-current, valve-current and thermal limits. The reactance order therefore passes through limiters and an inverse characteristic before becoming synchronized firing pulses.

Figure 7.22 illustrates oscillation damping on a single-machine infinite-bus system. Without modulation, an 80 ms fault leaves a sustained swing in the undamped machine model. With the TCSC reactance modulated in proportion to speed deviation, within ± 0.15 pu around its scheduled value, the oscillation is suppressed within about one second.

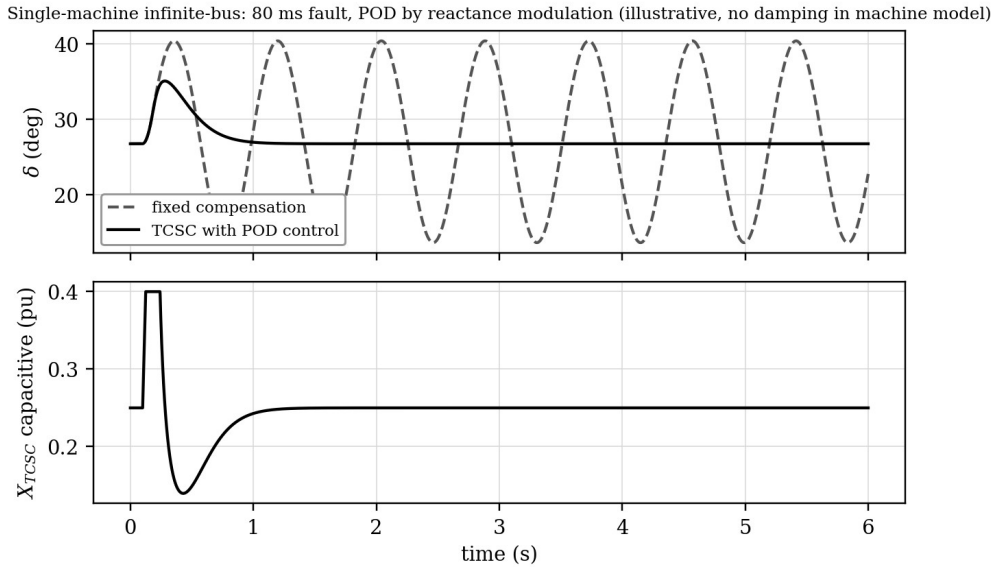


Figure 7.22. Rotor angle after an 80 ms fault with fixed compensation and with TCSC reactance modulation (illustrative SMIB model).

A power-oscillation damping channel is supplementary to the main power-flow loop. A local signal such as line power or current, or a remote synchrophasor signal, is filtered, passed through washout and phase-compensation

blocks, limited, and then added to the scheduled reactance order. Phase compensation is selected so that the electrical torque produced by the TCSC opposes the speed deviation over the target mode. Gain alone cannot guarantee positive damping because measurement filters, communication delay, firing dynamics and the network operating point all contribute phase shift.

Example 7.5. Supplementary damping command superimposed on TCSC reactance control

A TCSC operates at a steady capacitive reactance order of 12Ω . A power-oscillation damping controller adds a sinusoidal modulation of $\pm 1.5 \Omega$. State the commanded range and the checks required before the modulation is applied.

Solution. The order varies between 10.5 and 13.5 Ω . The internal controller must map this range into admissible firing instants while respecting the capacitive operating band, the resonance exclusion margin, and the capacitor-voltage, reactor-current and valve thermal limits. The damping channel must also keep its intended phase compensation over the range of modal frequencies.

Note. A damping controller is not validated by the amplitude of its reactance order. Positive damping must be demonstrated for the actual network, measurement filters, firing logic and operating points.

7.4.9 Subsynchronous behaviour and the NGH principle

Hingorani's NGH damper is a thyristor-controlled resistor across a series capacitor. It forces the capacitor voltage to zero at the end of each half-period of the system frequency, whenever the half-period is longer than that of the fundamental. A subsynchronous component, which lengthens the half-periods, is thereby dissipated. The TCSC firing that produces capacitive boost also reverses the capacitor voltage near the end of each half-cycle, so the circuits are strikingly similar. If the reversals are kept at equal intervals corresponding to the *fundamental* frequency, as the capacitor-voltage-synchronized control of Section 7.4.8 does, the TCSC presents an inductive, resistive impedance to subsynchronous currents while remaining capacitive at 50 Hz. Simulation studies and the Stöde experience confirm this behaviour. The property depends on the control, however: a TCSC synchronized to a poorly filtered line-current PLL can still interact with torsional modes. SSR screening is therefore always required.

7.4.10 Representation in power-flow and stability studies

For steady-state studies the TCSC is represented as a variable series reactance X_{TCSC} in the branch admittance matrix. With the line reactance X , the branch admittance becomes $1/[j(X + X_{TCSC})]$, and X_{TCSC} (or β) is added as a state variable regulated to a power-flow target. An equivalent *power-injection model* replaces the TCSC by the active and reactive powers it injects at the two end buses, which leaves the admittance matrix unchanged. For stability studies a first-order lag between the reactance order and the actual reactance, with a time constant of about one cycle, is usual. Example 7.37 works through the admittance form.

No single TCSC model is appropriate for every study. A variable-reactance model is suitable for load-flow scheduling and many electromechanical studies. A first-order dynamic reactance model can represent dominant command response in transient-stability simulations. SSR, harmonic, protection and switching studies require a waveform or EMT representation containing the capacitor, reactor, thyristor firing logic, MOV and bypass path. Dynamic-phasor and frequency-dependent models provide an intermediate level when subsynchronous behaviour must be retained without the full cost of switching simulation.

Model fidelity should be chosen from the phenomenon being studied. A damping controller that appears satisfactory with a first-order reactance model can change after current synchronization, resonance-exclusion logic and firing dynamics are included. Conversely, a detailed EMT model is unnecessary for an optimization that only requires the steady-state branch reactance.

7.4.11 Ratings and protection

Ratings follow from the capacitive boost at rated current, the maximum capacitor voltage (including harmonics), the thyristor RMS and peak current, and the transient overload profile, which for series capacitors typically allows of the order of 1.35 to 1.5 times rated current for periods of minutes to tens of minutes (the exact profile is set by the purchaser's specification). The MOV protects the capacitor and valve against fault overvoltages. Because the valve can bypass the module within a cycle, the TCSC can reduce MOV energy duty compared with a fixed capacitor, and its inductive vernier can limit fault current.

During severe faults, normal vernier control is subordinated to protection. Depending on the design, the thyristor valve is driven to bypass, the bypass breaker may close, and the MOV absorbs part of the capacitor energy. Recovery after fault clearing must avoid an uncontrolled transition through the resonant region. Modern TCSC specifications therefore treat operating-mode transitions, protection coordination and real-time control-system validation as part of the controller design.

7.5 The Static Synchronous Series Compensator (SSSC)

7.5.1 Definition and circuit

IEEE and CIGRE define the SSSC as a static synchronous generator operated without an external energy source as a series compensator, whose output voltage is in quadrature with, and controllable independently of, the line current, so as to increase or decrease the overall reactive voltage drop across the line and thereby control the transmitted power. The SSSC may include energy storage or a power supply on its DC side, in which case it can also exchange active power with the line.

Physically, an SSSC is a voltage-source converter, multi-pulse or multilevel, connected to the line through a series coupling transformer (Figure 7.23). The converter is a synchronous voltage source like the STATCOM of Chapter 6, but it is synchronized to the *line current* rather than to a bus voltage. Proposed by Gyugyi in 1989 [4], it has been realized in the converter-based installations of Inez and Marcy in the United States, where the series converter of a UPFC or CSC can be operated as an SSSC, and more recently in transformerless *modular* SSSCs that hang directly on transmission conductors.

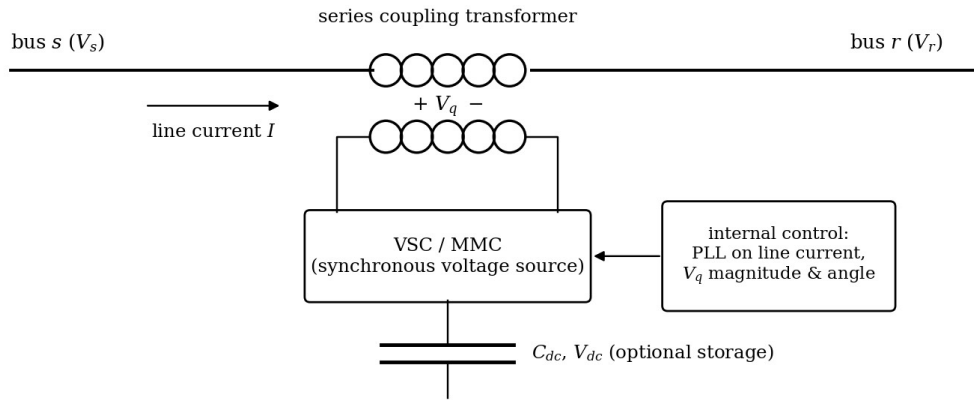


Figure 7.23. SSSC: voltage-source converter coupled to the line through a series transformer.

7.5.2 Power-angle characteristic

Refer once more to the symmetric two-machine system, now with the SSSC injecting a voltage V_q of magnitude V_q that lags the line current by 90° (capacitive mode, Figure 7.2b). This voltage adds directly to the end-voltage difference across the line reactance, so the current magnitude is

$$I = \frac{2V \sin(\delta/2) + V_q}{X} \quad (7.19)$$

where V_q is the injected quadrature voltage, positive for capacitive and negative for inductive operation. The current is in phase with the midpoint voltage $V \cos(\delta/2)$, so the transmitted power is $P = V_m I$:

$$P_q = \frac{V^2}{X} \sin \delta + \frac{V V_q}{X} \cos \frac{\delta}{2} \quad (7.20)$$

where P_q is the power transmitted with the SSSC. Compare this with the series capacitor, $P_c = V^2 \sin \delta / [(1 - k)X]$. The capacitor raises power by a fixed *fraction of the power at the same angle*, so it adds nothing at $\delta = 0$ and contributes most near 90° . The SSSC adds a term proportional to $\cos(\delta/2)$, a fixed *fraction*

of the maximum power V^2/X . This term is largest at small angles and does not vanish at $\delta=0$. Figure 7.24 shows both families. Two features follow directly from Equation (7.20).

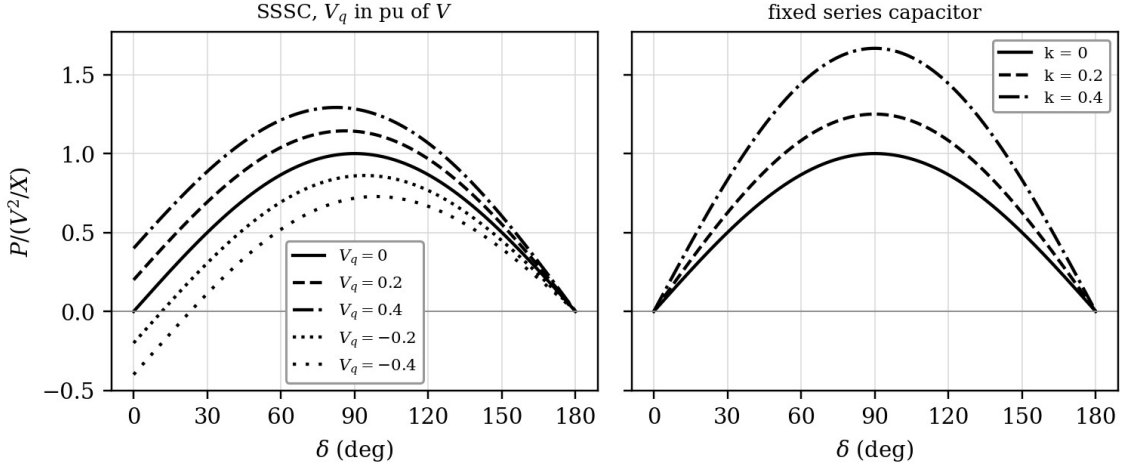


Figure 7.24. Normalized power-angle curves with an SSSC injecting V_q (left) and with a fixed series capacitor (right).

Symmetric range. The converter can produce $+V_q$ and $-V_q$ with the same rating, so an SSSC can *decrease* power flow as easily as it increases it. For power-flow control, its effective range is twice its VA rating, whereas a capacitor can only increase flow.

Power reversal. In inductive mode with $|V_q| > 2V \sin(\delta/2)$, the current, and with it the power, reverses direction. For $V_q = -0.5$ pu on a line of $X = 1.2$ pu, for example, the power becomes negative at angles below 29° (Example 7.39).

7.5.3 Operating modes and characteristics

In the *voltage compensating mode*, the SSSC maintains the rated compensating voltage, capacitive or inductive, at any line current up to I_{max} . Its VA rating is simply $V_{qmax} I_{max}$. In the *impedance compensating mode*, it emulates a reactance by making V_q proportional to current, $V_q = X_q I$, up to the voltage limit. Figure 7.25 shows both V-I areas. Unlike the capacitor-based devices, the SSSC can produce its full compensating voltage at low line current, which keeps power-flow control effective under light load. Its losses are those of a converter and transformer: roughly constant switching losses plus conduction losses that grow with current, so they are higher at light load than those of a TCSC.

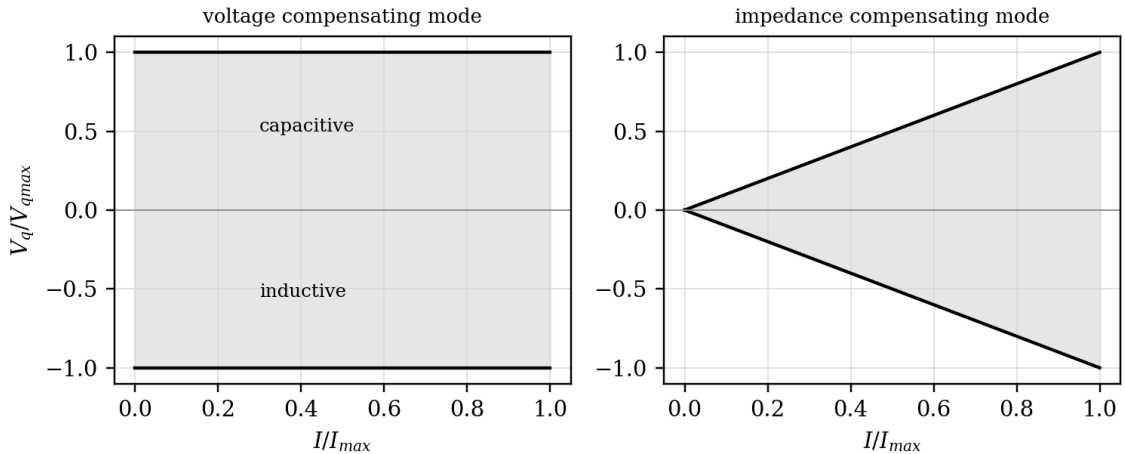


Figure 7.25. SSSC V-I characteristics: voltage compensating mode (left) and impedance compensating mode (right).

Example 7.6. SSSC constant-voltage versus constant-reactance operation

An SSSC can inject up to 20 kV per phase in quadrature with the line current. Compare the injected voltage at line currents of 0.5 kA and 1.0 kA when it operates in constant-voltage mode and when it emulates a series reactance of 20Ω .

Solution. In constant-voltage mode the order is 20 kV at both currents. In constant-reactance mode $V_q = I X_q$, giving 10 kV at 0.5 kA and 20 kV at 1.0 kA, where the converter reaches its voltage limit. Above 1.0 kA the emulated reactance can no longer be held.

Note. At very low current a constant-voltage order corresponds to a very large equivalent reactance V_q/I , so practical controllers use current thresholds and mode logic rather than extrapolating through zero current.

For balanced operation the converter rating is $S_{SSC} = \sqrt{3} V_{q,LL} I$, where $V_{q,LL}$ is the injected line-to-line RMS voltage and I the line RMS current. This is a converter MVA rating, not a capacitor kvar rating. In constant-voltage mode the MVA demand grows in proportion to the line current. In constant-reactance mode $V_q = X_q I$, so the demand grows with I^2 until the voltage ceiling is reached; beyond that point the controller leaves the ideal constant-reactance characteristic and operates on its voltage limit. The same converter therefore follows different V-I trajectories under the two commands, a distinction that matters in both rating and power-flow studies.

7.5.4 Real power exchange and compensation of line resistance

With resistance R , the power delivered to the receiving end of an uncompensated line is

$$P = \frac{V^2}{X^2 + R^2} [X \sin \delta - R(1 - \cos \delta)] \quad (7.21)$$

where R and X are the series resistance and reactance. Figure 7.26 shows how the resistive term erodes transfer capability as X/R falls. Reactive series compensation alone makes this worse, because reducing X lowers the effective X/R ratio. An SSSC with an energy source or sink on its DC side can inject a voltage component V_p in phase with the current in addition to V_q . That component cancels part of the resistive drop: the $I^2 R$ loss is still dissipated in the line, but the SSSC supplies it. The line then behaves as one with resistance $R - V_p/I$ and reactance $X - V_q/I$, and the X/R ratio can be kept high while the reactance is reduced (Example 7.41). Combined active and reactive modulation is also the most effective way to damp oscillations: maximum capacitive voltage with active power absorption while the machine accelerates, and the reverse while it decelerates.

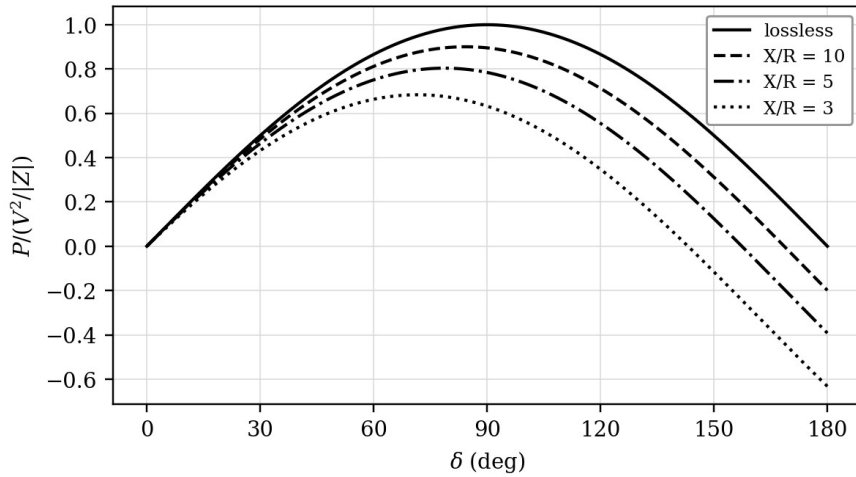


Figure 7.26. Normalized power-angle curves of a line with resistance for several X/R ratios.

Example 7.7. Active-power requirement for series resistance compensation

A line carries 800 A and has a series resistance of 1.5Ω per phase. Find the real power an energy-supported SSSC must supply to cancel the resistive voltage drop completely.

Solution. The in-phase voltage needed is $V_p = IR = 1.2$ kV per phase, and the power supplied is $3 V_p I = 3 I^2 R = 3 \times 800^2 \times 1.5 = 2.88$ MW.

Note. This exchange is continuous, so it requires an external energy source or a power path to the DC side; a DC capacitor alone cannot sustain it.

7.5.5 Converter, transformer and protection

The converter is built like a STATCOM converter: early designs used multi-pulse two-level or three-level bridges with harmonic-neutralizing series transformers, and new designs use cascaded H-bridge or MMC topologies. The series transformer must carry full line current, including fault current, and its secondary is protected by a thyristor bypass switch and a breaker that short the winding during faults. The converter is thereby isolated from fault currents it could not interrupt. A transformerless variant floats the converter at line potential, as in the modular SSSC units now installed in several networks.

The converter cannot carry the prospective line fault current, so the fast thyristor bypass and the mechanical bypass breaker are an essential part of its protection, and the series transformer must survive the fault current until they operate. Transformerless modular designs remove the series transformer but not the need for fault bypass, insulation coordination and protection-interaction studies (Section 7.7.2).

Multilevel converters synthesize the injected voltage from many smaller voltage steps, reducing harmonic content and semiconductor voltage stress. Cascaded and modular structures can also provide redundancy because a failed module may be bypassed while the remaining modules continue at reduced capability.

A modular SSSC distributes the required series voltage among several converter modules. Instead of one large substation converter and series transformer, smaller controllable voltage sources can be deployed at selected locations and expanded or relocated as the network changes. The control problem then becomes coordinated: modules must share voltage orders, respect local current and thermal limits, and respond coherently to faults and protection commands.

7.5.6 Immunity to series resonance and SSR

A capacitor's impedance depends on frequency, which is the root of series resonance. An ideal voltage-source converter with a constant DC voltage instead produces a voltage only at the frequency it is commanded to produce, and its output impedance is that of the coupling transformer leakage, which is inductive. An SSSC therefore cannot be tuned into classical series resonance with any finite line inductance, and its compensating voltage is unaffected by changes in network impedance. In practice the DC side is a finite capacitor. A subsynchronous line-current component at $f_0 - f_m$ produces, through the switching, an AC component of frequency f_m on the DC side, which in turn modulates the output voltage at $f_0 \pm f_m$. If the DC capacitance is large enough to keep this ripple small, or if the control keeps the output voltage vector exactly in quadrature with the fundamental current, the SSSC remains neutral to SSR. With an active damping controller it can even damp existing subsynchronous oscillations, although that technique is less established than TCSC-based damping.

The absence of classical series resonance must not be read as immunity from every subsynchronous phenomenon. The converter, its synchronization loop, DC-link dynamics, modulation and limiters, together with the surrounding network, form a controlled dynamic system. That system can take part in subsynchronous or supersynchronous control interactions even though no physical capacitor reactance crosses the line inductive reactance. Section 7.7.3 develops the impedance-based analysis used to screen such interactions.

7.5.7 Control

The internal control of an SSSC (Figure 7.27) synchronizes to the line current through a PLL. It constructs the output voltage vector from a quadrature component V_q , demanded by the external control, and a small in-phase component V_p , demanded by the DC-voltage regulator to cover losses or by the external control when storage is available. With an *indirectly controlled* converter, the DC voltage is varied through a transient angle shift, as in the STATCOM of Chapter 6; with a *directly controlled* converter, the voltage magnitude is set by modulation at constant DC voltage. The external control regulates line power, current or effective impedance, and adds damping and transient-stability functions.

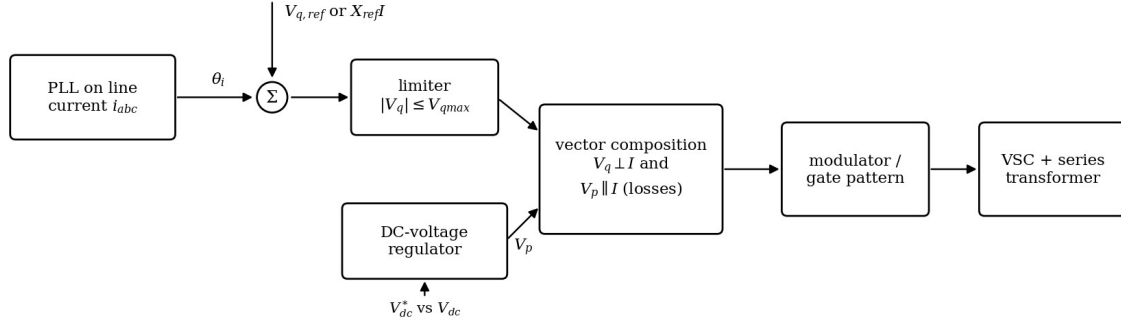


Figure 7.27. SSSC control structure.

In both schemes the power-system controller should be kept separate from the fast converter controller. The former requests a series voltage, a power flow or an equivalent reactance; the latter produces a feasible voltage vector within modulation, current, DC-voltage and semiconductor limits. With indirect control, the switching pattern and the DC-link voltage set the synthesized AC voltage, and a small phase displacement from exact quadrature supplies the converter losses and regulates the DC-link energy. With direct control, PWM or multilevel modulation sets the output-voltage vector explicitly.

For a purely reactive SSSC without external storage, the steady-state injected voltage is maintained approximately in quadrature with fundamental line current. Average real power exchanged with the AC system is then only that required to cover losses. A small in-phase voltage component regulates the DC link. If an energy source is connected to the DC side, the in-phase component can become an intentional control variable and sustained active-power exchange becomes possible subject to converter MVA and stored-energy limits.

For steady power-flow control, the SSSC adjusts the sign and magnitude of quadrature voltage so that measured line power or current follows its reference. For damping, a supplementary controller modulates the voltage order with a small signal derived from local or remote measurements. Because the converter can reverse the sign of series voltage without crossing a capacitor resonance, it can move smoothly between capacitive and inductive action. During a large disturbance, a transient-stability controller may temporarily request maximum capacitive injection to raise electrical power during the first swing, provided current and voltage limits permit it.

These functions compete for the same converter capability. Practical priority logic allocates available series voltage among the steady power-flow order, damping signal, DC-voltage requirement and emergency support. If storage is present, active-power exchange introduces an additional energy constraint. A complete SSSC model therefore needs both an AC-side converter limit and a DC-side energy balance.

7.5.8 Hybrid schemes

A fixed series capacitor combined with an SSSC (for example 0.5 pu of each) is an economical way to obtain a large compensation level with controllability: the capacitor provides cheap reactance and the SSSC provides fine control, damping and SSR mitigation. The SSSC can be controlled to cancel the subsynchronous voltage that the fixed capacitor would produce, which is the principle behind SSR mitigation studies on the IEEE first benchmark model. The hybrid also extends the attainable V-I area beyond that of either element alone (Example 7.43).

7.6 Comparison and Selection

Figure 7.28 contrasts the compensating voltage of the main series devices as a function of line current. A fixed capacitor or an inserted TCSC stage produces a voltage proportional to current. A TCSC can boost that voltage at low current but is eventually limited by its capacitor voltage rating. An SSSC produces a voltage independent of current. Table 7.3 collects the comparison.

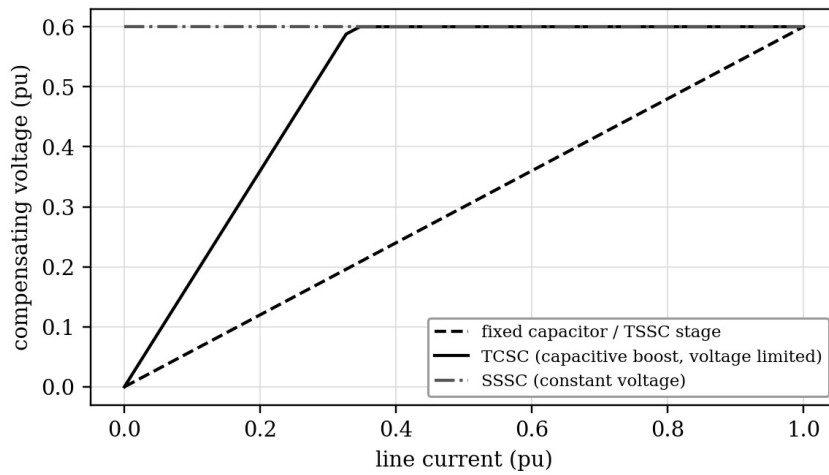


Figure 7.28. Compensating voltage versus line current for a capacitor (or TSSC stage), a TCSC and an SSSC of similar voltage rating.

Table 7.3 Comparison of series FACTS controllers.

Attribute	GCSC	TSSC	TCSC	SSSC
Control	Continuous, capacitive only	Stepped, capacitive	Continuous, capacitive boost and inductive	Continuous, capacitive and inductive, symmetric
Switching devices	GTO/IGCT (turn-off)	Thyristors	Thyristors	IGBT/IGCT converter
Harmonics	Voltage harmonics (TCR dual)	None in steady state	Voltage harmonics, small at modest boost	Low with multilevel converter
Output at low current	Falls with current	Falls with current	Boost possible, falls eventually	Independent of current
SSR behaviour	Capacitive (control dependent)	Same as fixed capacitor	Inductive at subsynchronous frequencies with suitable control	No series resonance
Power reversal	No	No	No (inductive vernier only reduces flow)	Yes
Active power	No	No	No	Yes, with DC storage
Response	Half cycle	Half to one cycle	About one cycle	Sub-cycle current-loop dependent
Installation	Platform at line potential	Platform	Platform	Series transformer and ground-level converter, or modular at line potential
Maturity	Concept and distribution use	Limited	Widely installed	Few transmission units; modular SSSC growing

The choice among them depends on what the compensation must achieve. For bulk transfer increase over long lines, fixed capacitors remain the cheapest per megavar, with a TCSC module added where damping or SSR control is needed. For power-flow control in meshed networks, where the flow may need to be reduced as well as increased, the SSSC's symmetric range and power-reversal ability are decisive. Where only coarse scheduling is required and SSR is not a concern, a TSSC or mechanically switched capacitor is adequate. Thyristor-controlled series reactors (TCSR) and thyristor-switched series reactors (TSSR), which control a series inductor instead of a capacitor, are mainly used for fault-current limiting and are treated with phase-angle regulators in Chapter 13.

Selection should be based on the required service rather than on a simple ranking, and Table 7.4 reorganizes the comparison in that way. The decisive questions are whether compensation must be continuous or stepped, whether the controller must both increase and decrease corridor flow, whether a classical series capacitor is acceptable from the SSR perspective, how much fault-current duty the series equipment must survive, and whether converter losses and protection complexity are justified by the additional controllability.

Table 7.4 Service-oriented comparison of the four series FACTS controller families.

Attribute	GCSC	TSSC	TCSC	SSSC	Engineering implication
Control resolution	Continuous capacitive	Discrete	Continuous with capacitive and inductive regions	Continuous voltage of either sign	Match resolution to flow-control and damping duty
Classical capacitor resonance	Present	Present	Present, with control-dependent behaviour	No literal fixed-capacitor resonance	Still assess converter-control interactions
Low-current behaviour	Compensating voltage falls with current	Compensating voltage falls with current	Operating range depends on capacitor and reactor limits	Voltage-source command retained above minimum-current logic	Useful for bidirectional flow routing
Fault strategy	Fast bypass	Valve and breaker bypass	MOV, valve and bypass coordination	Converter bypass and relay coordination	Use EMT and protection studies
Key research issue	Stress and harmonics	Stage coordination	Dynamic model, SSR/SSCI and POD	Modularity, protection and converter-grid dynamics	Validate control and protection together

7.7 Recent Advances, Research Gaps and Emerging Directions

The physics developed in Sections 7.1 to 7.5 has not changed, but the setting in which series compensation is applied has. Converter-interfaced generation now dominates many corridors, series converters are increasingly deployed as fleets of small modules rather than single substation installations, and validation has moved from fundamental-frequency studies towards electromagnetic transient (EMT) simulation and hardware-in-the-loop testing. Three threads run through the recent literature: series compensation used as a *power-flow-routing actuator*, the *wide-band interaction* between converter controls and series-compensated networks, and *protection-aware design and validation*. This section develops each thread with the equations a research scholar needs to begin quantitative work, and closes with a set of open problems. Table 7.5 summarizes the gaps before they are examined in turn.

Table 7.5 Research gaps and possible contributions in series compensation.

Theme	Limitation of current practice	Potential contribution	Measurable impact
TCSC design models	Simplified LC model misplaces resonance	Exact or dynamic-phasor models in design tools	Correct mode and resonance margin
SSR and SSCI	Screening relies on case-by-case EMT studies	Impedance-based design of TCSC/SSSC damping	Positive net damping across 5 to 45 Hz
Modular SSSC fleets	Units dispatched individually	Optimal placement and coordinated dispatch with N-1 constraints	Congestion relief per installed MVA
Protection interaction	Converters often modelled as ideal shorts during faults	Bypass-aware relay studies and adaptive settings	Zone-1 security and dependability margins
Wide-area damping	Delay and loss of signals	Delay-robust, adaptive damping controllers	Damping ratio over the expected delay range

7.7.1 Modular and distributed series compensation for power-flow routing

A modular static synchronous series compensator (M-SSSC) distributes the series voltage of Section 7.5 among N small converter modules, each with its own bypass, mounted on the conductor or on a small platform. Field deployments now include a cross-border corridor of the Greek transmission system, where M-SSSC units are used to raise interconnection capacity and the integration of renewable generation [17], and several 220 kV circuits in Colombia, where operating experience with parallel lines has been reported [15]. Modularity changes the design question. The question is no longer only how far one line can be pushed, but how a fleet of voltage sources should be placed and dispatched in a meshed network.

If the active modules inject fundamental voltages aligned with a common reference, their contributions add:

$$V_{\Sigma} = \sum_{k \in A} V_k, |V_k| \leq V_{k,max} \quad (7.22)$$

where V_{Σ} is the aggregate series voltage, A the set of available (non-bypassed) modules and $V_{k,max}$ the voltage rating of module k . A useful fleet metric is the headroom

$$h = 1 - \frac{|V_{\Sigma}^*|}{\sum_{k \in A} V_{k,max}} \quad (7.23)$$

where V_{Σ}^* is the aggregate voltage order. With equal sharing each module supplies $|V_{\Sigma}^*|/|A|$, so every bypassed module raises the duty of the survivors. Figure 7.29(b) shows a six-module fleet of 8 kV units asked for 24 kV: three module outages consume all the headroom (Example 7.8 gives a four-module case).

For routing, it is convenient to express the injection as an emulated reactance $X_{inj} = V_q/I$ (positive when inductive). For two parallel paths of reactance X_1 and X_2 carrying a total P_t , with the series device in path 1,

$$P_1 = P_t \frac{X_2}{X_1 + X_{inj} + X_2}, \frac{\partial P_1}{\partial X_{inj}} = - \frac{P_t X_2}{(X_1 + X_{inj} + X_2)^2} \quad (7.24)$$

where P_1 is the flow in the compensated path. Inductive injection pushes power out of the compensated path and capacitive injection draws power into it (Figure 7.29a). In the example of the figure ($X_1 = 0.3$ pu, $X_2 = 0.4$ pu, $P_t = 1.6$ pu), line 1 naturally carries 0.914 pu against a thermal limit of 0.85 pu. An inductive emulated reactance of 0.053 pu removes the overload. At a line current of about 0.85 pu this needs only $V_q \approx 0.045$ pu of series voltage, which shows why small, distributed converters can relieve congestion so effectively.

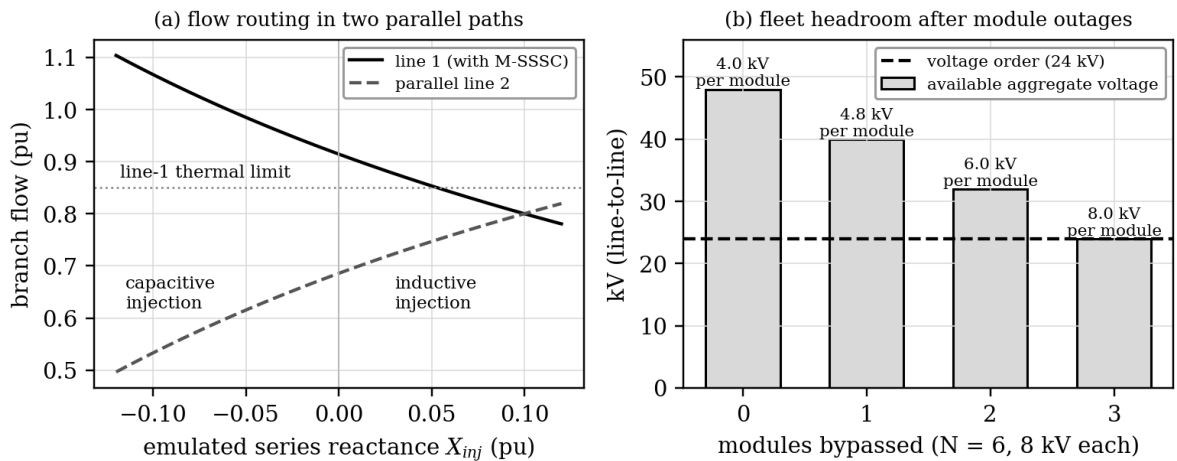


Figure 7.29. Modular series compensation: (a) redistribution of flow between two parallel paths as the emulated series reactance of path 1 is varied; (b) aggregate voltage available from a six-module fleet as modules are bypassed, with the voltage each survivor must supply.

At system scale, the fleet is dispatched by an optimal power flow in which the branch equation of each compensated line contains the aggregate injection:

$$V_s - V_r = Z_\ell I + V_\Sigma, |I_\ell^{(c)}| \leq I_{\ell, \max} \quad \forall \ell, \forall c \in C \quad (7.25)$$

where Z_ℓ is the line impedance, $I_\ell^{(c)}$ the current of line ℓ in contingency c of the set C , and $I_{\ell, \max}$ its thermal limit. The objective may be production cost, curtailment of renewable generation, or the redispatch needed to respect the limits. Formulated this way, the value of modular series control can be compared directly with phase-shifting transformers, topology switching, dynamic line rating and new lines [17]. Placement and sizing are then integer decisions layered on top of this continuous dispatch problem.

7.7.2 Interaction with line protection

A series device sits inside the measurement loop of the distance relays protecting its line. For a fault at a fraction m of the line beyond the device, the relay voltage contains the injected voltage, and the apparent impedance measured at the relay is

$$Z_{app} = \frac{V_{relay}}{I} = m Z_L + \frac{V_{inj}}{I} \quad (7.26)$$

where Z_L is the total line impedance and V_{inj} the series voltage present during the fault. For quadrature injection, $V_{inj}/I = \mp j X_q$. Capacitive injection ($-j X_q$) makes the fault look closer and causes *overreach*; inductive injection ($+j X_q$) makes it look farther and causes *underreach*. Figure 7.30 shows both effects against a zone-1 mho characteristic set at 80% of the line. With $X_q = 0.08$ pu, a fault at 85% of the line moves inside zone 1 under capacitive injection, and a fault at 70% moves outside it under inductive injection (Example 7.10).

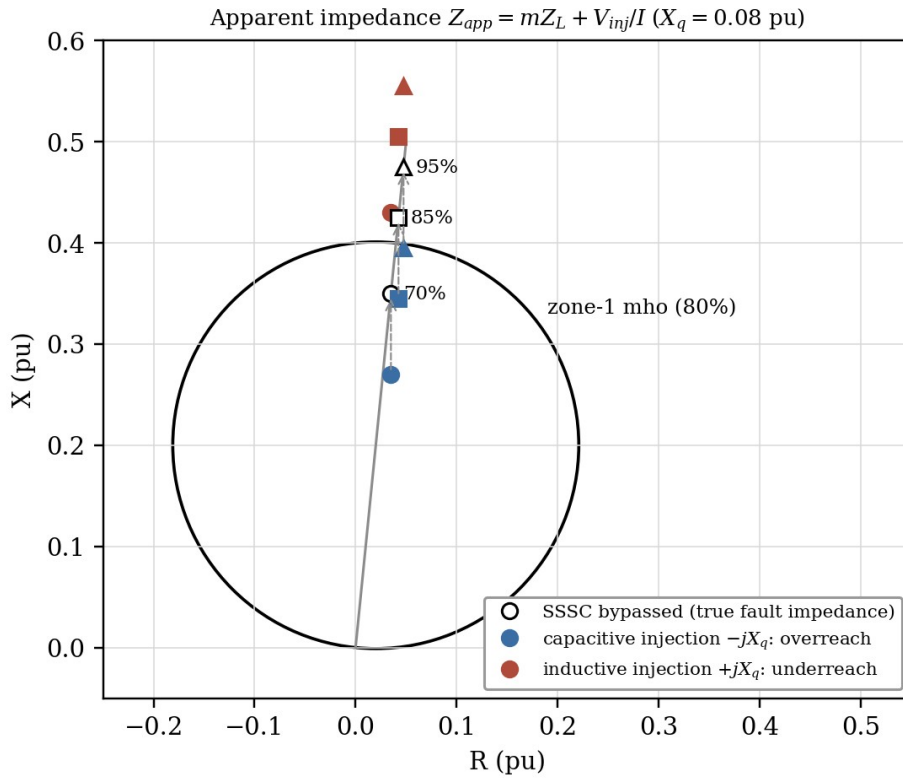


Figure 7.30. Apparent impedance seen by a distance relay for faults at 70%, 85% and 95% of a line ($Z_L = 0.05 + j0.5$ pu): true fault impedance (SSSC bypassed) and the shifts produced by capacitive and inductive series injection of 0.08 pu.

Equation (7.26) also explains why bypass timing matters. The converter does not vanish at the fault instant. It first passes through current limiting, a possible loss of synchronization and blocking before the bypass closes, so the apparent impedance follows a time-varying trajectory during the relay's measurement window. Recent field and laboratory studies of transmission SSSC projects therefore reproduce the actual bypass logic in EMT models and test relays with protection hardware-in-the-loop [15], [16]. A research model that replaces the converter by an instantaneous short circuit cannot establish zone-1 security.

7.7.3 SSR, SSCI and impedance-based stability analysis

Classical SSR (Section 7.1.5) couples an electrical series resonance with a turbine-generator torsional mode. Two further mechanisms dominate recent work. The *induction-generator effect* arises because a machine rotor turning near synchronous speed presents a negative resistance to subsynchronous stator currents. *Subsynchronous control interaction* (SSCI) arises when the current controllers of converter-interfaced generation, notably doubly fed induction generators, add further negative resistance at subsynchronous frequencies. SSCI involves no mechanical mode at all, and it has caused damaging oscillations near series capacitors in several networks [19]. Grid-forming inverters connected radially through series capacitors can also exhibit SSR, so the problem does not disappear with the change of control paradigm [18].

A first-order screening criterion follows from the series resonant circuit. At the electrical resonance $f_e = f_0 \sqrt{k}$ (Equation (7.5)) the network reactance vanishes, and the oscillation is damped only if the net series resistance at that frequency is positive:

$$R_{net}(f_e) = R_\ell + R_s + \frac{R_r' f_e}{f_e - f_0} + R_c(f_e) > 0 \quad (7.27)$$

where R_ℓ is the line resistance, R_s and R_r' the stator and referred rotor resistances of the machine, f_0 the system frequency, and $R_c(f_e)$ the resistance contributed by converter control (negative in the critical band). The term $R_r' f_e / (f_e - f_0)$ is negative because the slip of the rotor relative to a subsynchronous field is negative, and it grows in magnitude as f_e approaches f_0 , that is, as compensation increases. Figure 7.31 evaluates Equation (7.27) for representative per-unit data. With the induction-generator effect alone, the net resistance crosses zero at $k = 64\%$. A modest converter contribution moves the boundary down to about 59% (Example 7.11).

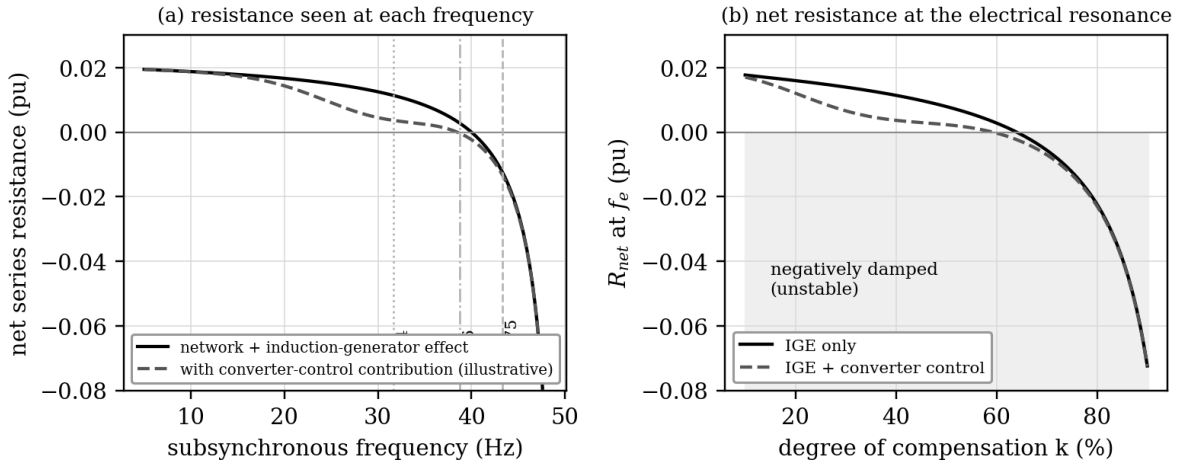


Figure 7.31. Screening of subsynchronous instability: (a) net series resistance versus frequency, with the electrical resonance for several compensation levels; (b) net resistance at the resonance versus compensation level ($R_\ell + R_s = 0.02$ pu, $R_r' = 0.005$ pu; converter contribution illustrative).

For converter-dominated systems the scalar criterion generalizes to an impedance-ratio test. The grid and the converter are linearized around an operating point, and their terminal impedances $Z_g(s)$ and $Z_c(s)$ are obtained in the sequence or dq frame. The interconnection is stable if the minor-loop gain

$$L(s) = Z_g(s) Z_c^{-1}(s) \quad (7.28)$$

satisfies the generalized Nyquist criterion [20]. For three-phase converters with PLLs, Z_c is a 2×2 matrix, and the scalar screening of Equation (7.27) is only a first pass. In a series-compensated network $Z_g(s)$ contains the term $1/(sC)$ of the capacitor, which produces the low-impedance resonance that makes such corridors particularly sensitive. Candidate frequencies found by impedance scans must be confirmed by eigenvalue analysis or EMT disturbance tests, because limiters and switching logic are not captured by the linear model.

7.7.4 Wide-area damping and communication delay

A series controller damps an inter-area mode most effectively when it sits on a corridor with high modal observability and controllability, and remote synchrophasor signals often observe the mode better than local ones [21]. The price is communication delay. In the damping loop

$$\Delta V_q(s) = G_{POD}(s) e^{-sT_d} \Delta y(s) \quad (7.29)$$

where Δy is the measured oscillatory signal, G_{POD} the washout and phase-compensation network and T_d the end-to-end delay, the delay adds a phase lag that grows linearly with frequency:

$$\phi_d = -360^\circ f_m T_d, e^{-sT_d} \approx \frac{1 - sT_d/2}{1 + sT_d/2} \quad (7.30)$$

where f_m is the modal frequency; the second expression is the first-order Padé approximation used to include the delay in linear models. Delay compensation in G_{POD} restores the phase at one frequency only, and network conditions shift the modal frequency, so robust designs are evaluated over a range of delays and operating points [22].

The effect can be seen in a single-mode model. Representing an electromechanical mode of natural frequency $\omega_n = 2\pi f_m$ and open-loop damping ζ_0 , with the series controller providing speed-proportional damping of gain K through the delayed channel, the closed-loop characteristic equation is

$$s^2 + 2\zeta_0\omega_n s + \omega_n^2 + K s e^{-sT_d} = 0 \quad (7.31)$$

where K is chosen to give 15% damping without delay. Figure 7.32 shows the damping ratio of the dominant root as the delay increases. The 0.4 Hz mode tolerates a delay of 200 ms with little loss, whereas the 1.0 Hz mode falls below the common 5% criterion at about 190 ms and becomes unstable before 250 ms. Delay budgets must therefore be set per mode, and loss-of-signal fallback to a local signal is an essential part of the design (Examples 7.9 and 7.12).

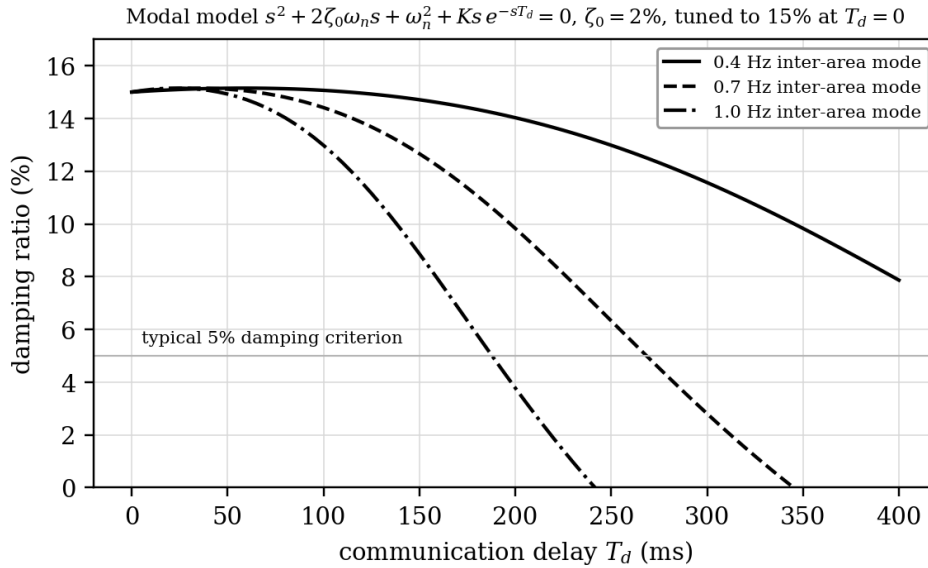


Figure 7.32. Damping ratio of an inter-area mode versus communication delay in the wide-area damping loop of Equation (7.31) ($\zeta_0 = 2\%$, gain tuned for 15% damping without delay).

7.7.5 Model fidelity and validation

No single model is adequate for every claim made about a series FACTS controller. Each level of the ladder in Figure 7.33 adds phenomena that the level below cannot represent. Table 7.6 maps the levels to the questions they can answer. The trend in standards follows the same direction: the 2023 edition of IEC 60143-4 replaces the earlier real-time network-simulator test of TCSC control and protection with a hardware-in-the-loop test [13]. For modular SSSCs, EMT models suitable for both offline and real-time studies are now published by the equipment community [14].

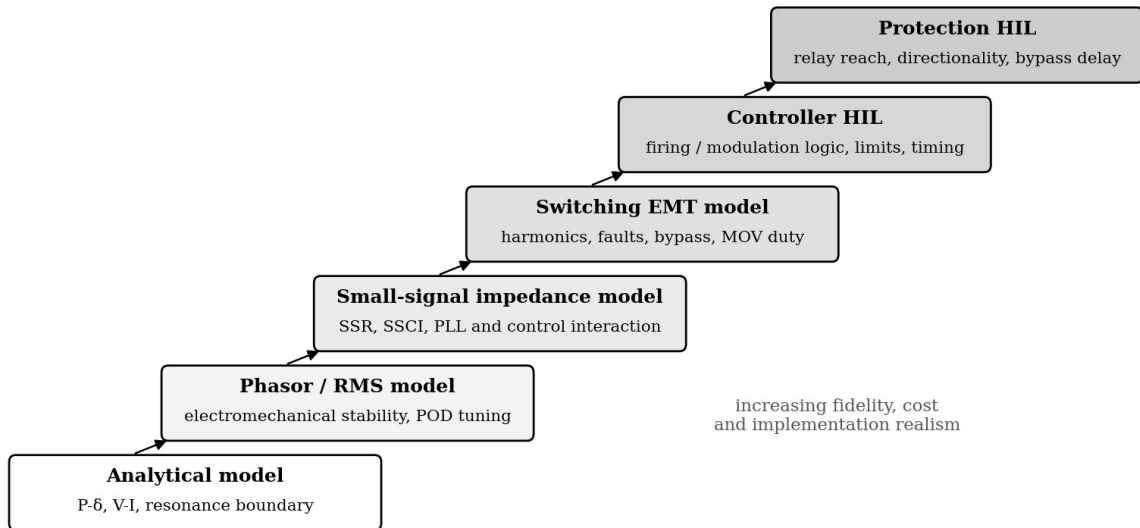


Figure 7.33. Validation ladder for series FACTS controllers, from analytical models to protection hardware-in-the-loop.

Table 7.6 Model fidelity and the claims each level can support.

Model level	Represents	Suitable claims	Cannot establish
Analytical / steady state	Fundamental-frequency phasors	Transfer capability, ratings, resonance boundary	Dynamics, harmonics, protection
Phasor (RMS) dynamic	Electromechanical dynamics, averaged controls	POD tuning, transient stability	SSR/SSCI, switching events
Small-signal impedance	Linearized converter and network to about 1 kHz	SSR, SSCI, PLL interaction screening	Limiter action, large disturbances
Switching EMT	Valves, firing or modulation, MOV, bypass	Harmonics, fault recovery, protection duty	Behaviour of the real controller code
Controller HIL	Real control hardware with simulated plant	Timing, mode logic, limits, firmware	Relay response
Protection HIL	Real relays with simulated converter and line	Reach, directionality, bypass timing	Long-term planning metrics

7.7.6 Open research questions

The most productive research questions follow from limitations established earlier in this chapter. Table 7.7 lists representative questions together with a suggested method and a measurable outcome. A strong contribution identifies one such limitation, proposes a controller, model or topology that addresses it, and demonstrates the improvement over several operating conditions and at more than one level of the validation ladder.

Table 7.7 Representative research questions in series compensation.

Research question	Why it matters	Suggested method	Measurable outcome
Reduced-order TCSC models that keep the exact resonance boundary	Simplified models misplace resonance (Section 7.4.3)	Dynamic phasors, harmonic-domain models	Error in β_r and in subsynchronous impedance
Robust POD for series devices under changing grid strength and delay	Damping varies with T_d and operating point (Figure 7.32)	H_∞ , gain scheduling, data-driven tuning	Minimum damping ratio over the operating set
Coordinated dispatch of M-	Headroom is consumed by module outages (Figure 7.29b)	Security-constrained OPF with fleet constraints	Congestion relief per installed MVA

Research question	Why it matters	Suggested method	Measurable outcome
SSSC fleets with N-1 security			
Protection-aware series converter control	Injection shifts relay reach (Figure 7.30)	EMT and protection HIL, adaptive bypass	Zone-1 overreach and underreach margins
SSCI-robust converter and series-device control	Negative resistance near f_e (Figure 7.31)	Impedance shaping, passivity-based design	Net resistance or passivity margin across 5 to 45 Hz

Worked examples for Section 7.7

Example 7.8. Modular SSSC loss of one module

A four-module SSSC must inject 24 kV line-to-line. The modules share voltage equally and each is rated 8 kV. If one module is bypassed, determine the voltage required from each remaining module and state whether the original command remains feasible.

Solution. Initially each module supplies $24/4=6$ kV and the headroom from Equation (7.23) is $1 - 24/32=25\%$. With one module bypassed, the remaining three must each supply $24/3=8$ kV. The command is still feasible, but the headroom is now zero and the survivors operate exactly at their rating.

Note. A single module outage can consume all control headroom even when the aggregate command remains achievable, which is why fleet dispatch must include module availability.

Example 7.9. Communication delay in a wide-area POD loop

A remote damping signal contains a 0.7 Hz inter-area oscillation. Estimate the additional phase lag caused by communication delays of 50 ms and 150 ms.

Solution. From Equation (7.30), $\phi_d = -360 \times 0.7 \times 0.05 = -12.6^\circ$ for 50 ms and -37.8° for 150 ms.

Note. A delay that looks small in milliseconds can consume a large part of the phase compensation designed for an inter-area mode.

Example 7.10. Distance-relay reach with series injection

A line has $Z_L=0.05+j0.5$ pu and a zone-1 mho relay set at 80% of the line, with its characteristic passing through the origin. An SSSC between the relay and the line injects $X_q=0.08$ pu. Determine whether zone 1 operates for (a) a fault at 85% with capacitive injection and (b) a fault at 70% with inductive injection.

Solution. The mho circle has centre $0.201 \angle 84.3^\circ = 0.020 + j0.200$ pu and radius 0.201 pu. (a) $Z_{app} = 0.0425 + j0.425 - j0.08 = 0.0425 + j0.345$ pu, at a distance of 0.147 pu from the centre: inside zone 1, so the relay trips instantaneously for a fault beyond its reach (overreach). (b) $Z_{app} = 0.035 + j0.35 + j0.08 = 0.035 + j0.43$ pu, at 0.230 pu from the centre: outside zone 1, so a fault that zone 1 should clear is delayed to zone 2 (underreach).

Note. Both errors disappear once the bypass closes, so the relay measurement window and the bypass time must be coordinated.

Example 7.11. Compensation limit from the induction-generator effect

For $R_\ell + R_s = 0.02$ pu and $R_r' = 0.005$ pu at 50 Hz, find the compensation level at which the net resistance of Equation (7.27) becomes zero, ignoring converter control.

Solution. Setting $0.02 + 0.005 f_e / (f_e - 50) = 0$ gives $0.02(f_e - 50) = -0.005 f_e$, so $f_e = 40$ Hz and $k = (40/50)^2 = 0.64$. Compensation above 64% is negatively damped at the electrical resonance.

Note. Converter control can move this boundary considerably; with the illustrative contribution of Figure 7.31 it falls to about 59%.

Example 7.12. Delay tolerance of a wide-area damping loop

For the single-mode model of Equation (7.31) with $\zeta_0 = 2\%$ and the gain tuned for 15% damping without delay, find the damping ratio of a 0.7 Hz mode for delays of 150 ms and 200 ms.

Solution. Solving Equation (7.31) for the dominant root gives 12.7% at 150 ms and 9.8% at 200 ms (Figure 7.32). The corresponding phase lags from Equation (7.30) are 37.8° and 50.4° .

Note. Adding 37.8° of lead at 0.7 Hz restores the design damping at that frequency only; robustness to frequency drift must be checked separately.

Key Equations Recap

Table 7.8 Key equations of Chapter 7.

Equation	Expression	Conditions
(7.3)	$I = \frac{2V}{(1-k)X} \sin \frac{\delta}{2}, P = \frac{V^2}{(1-k)X} \sin \delta$	Symmetric lossless line, $k = X_C / X$
(7.4)	$Q_C = \frac{2V^2}{X} \frac{k}{(1-k)^2} (1 - \cos \delta)$	Series capacitor reactive power
(7.5)	$f_e = f_0 \sqrt{k}$	Electrical resonance; rotor sees $f_0 - f_e$
(7.6)	$v_C = \frac{I}{\omega C} (\sin \omega t - \sin \gamma)$	GCSC, $\gamma \leq \omega t \leq \pi - \gamma$
(7.8)	$X_C(\gamma) = X_C \left(1 - \frac{2\gamma}{\pi} - \frac{\sin 2\gamma}{\pi} \right)$	GCSC fundamental reactance
(7.9)	$V_{C,rms} = \frac{I}{\omega C} \sqrt{\frac{(\pi - 2\gamma) \left(\frac{1}{2} + \sin^2 \gamma \right) - \frac{3}{2} \sin 2\gamma}{\pi}}$	GCSC total RMS voltage
(7.11)	$v_C = I_m X_C (1 - \cos \omega t)$	TSSC insertion at current zero
(7.14)	$X_{TCSC} = \frac{X_C X_L(\alpha)}{X_C - X_L(\alpha)}$	Simplified model only
(7.16)	$X_{TCSC} = -X_C + C_1(2\beta + \sin 2\beta) - C_2 \cos^2 \beta (\omega \tan \omega \beta - \tan \beta)$	Exact, current-driven, β half conduction
(7.18)	$\beta_r = \pi / (2\omega)$	TCSC resonance (first)
(7.20)	$P_q = \frac{V^2}{X} \sin \delta + \frac{V V_q}{X} \cos \frac{\delta}{2}$	SSSC, lossless line
(7.21)	$P = \frac{V^2}{X^2 + R^2} [X \sin \delta - R(1 - \cos \delta)]$	Line with resistance

Common Misconceptions

“Series compensation controls power flow.” Fixed series compensation raises transfer capability, but the actual flow is still set by the angle and by the network. Control of flow requires a controllable device.

“The SSR frequency is f_e .” The electrical resonance is at $f_e = f_0 \sqrt{k}$; the rotor experiences the complement $f_0 - f_e$, and it is that frequency which must avoid the torsional modes.

“The simplified TCSC formula is good enough for design.” It misplaces the resonance and can put an operating point on the wrong side of it. Use the exact characteristic, Equation (7.16), or a validated time-domain model.

“A TCSC is immune to SSR.” It presents an inductive impedance at subsynchronous frequencies only when its firing is synchronized so that the charge reversals occur at fundamental-frequency intervals. Screening is still required.

“A TSSC can be inserted at any instant.” Thyristors commute only at current zero, which inserts the capacitor with a DC offset unless it is pre-charged. Bypassing should be done at capacitor voltage zero.

“The SSSC power term needs a unit conversion factor.” With line-to-line kilovolts and ohms, both V^2/X and $V V_q/X$ come out directly in megawatts for the three-phase system. An extra factor, as in some worked solutions, overstates the SSSC contribution tenfold (1108.7 MW instead of 110.9 MW in Example 7.40).

“An SSSC needs a large DC capacitor to supply reactive power.” As with the STATCOM, balanced reactive power circulates among the phases. The DC capacitor handles ripple and losses; only active power exchange needs storage.

Chapter Summary

Series compensation acts on the transfer reactance, which makes it the most effective means of raising the steady-state and transient transfer limits of long lines. Its price is the subsynchronous resonance that a series capacitor forms with the network inductance. The GCSC, the dual of the TCR, controls the fundamental capacitor voltage through a turn-off delay. The TSSC switches stages with natural commutation, which forces insertion at current zero and creates DC offsets. The TCSC tunes a parallel LC circuit; its true behaviour is governed by capacitor charge reversal under an imposed line current. The exact reactance law differs markedly from the simplified voltage-source model, and correctly synchronized, it makes the TCSC inductive at subsynchronous frequencies. The SSSC injects a quadrature voltage independent of line current, offers a symmetric range, can reverse power, cannot form a series resonance, and with storage can compensate line resistance as well as reactance.

Solved Numerical Practice Problems

Unless stated otherwise the system frequency is 50 Hz ($\omega=314.159$ rad/s), line-to-line voltages and ohmic reactances give three-phase powers directly ($P=V_{LL}^2 \sin \delta / X$), and positive reactance denotes inductive behaviour. For the TCSC, α is the firing delay measured from the crest of the capacitor voltage and $\beta=90^\circ - \alpha$ the half conduction angle. Several problems are adapted from classroom exercises and have been reworked for consistent sign conventions and units.

Example 7.13 Rating a series capacitor for a 50% power increase

A 400 kV line has $X=80 \Omega$ and equal end voltages and operates at $\delta=30^\circ$. Find (a) the line current; (b) the transmitted power; (c) the reactive power drawn by the line from each end; (d) the degree of compensation for a 50% increase in power at the same angle; (e) the new current; (f) the capacitor reactance and three-phase rating.

Solution. (a) $V_{ph}=230.94$ kV; $I=2 V_{ph} \sin 15^\circ / X=1494.3$ A.

(b) $P=400^2 \sin 30^\circ / 80=1000$ MW (three-phase).

(c) $Q=V^2(1 - \cos \delta) / X=267.9$ Mvar at each end.

(d) $1/(1 - k)=1.5$, so $k=1/3$.

(e) $I'=I/(1 - k)=2241.4$ A.

(f) $X_C=26.67 \Omega$; $Q_C=3 I'^2 X_C=401.9$ Mvar, which agrees with Equation (7.4).

Note. The capacitor must be rated for the boosted current, not the original one, which is why a 50% transfer gain at the same angle costs about 400 Mvar of series capacitors.

Example 7.14 Current, power and capacitor duty at 40% compensation

A 220 kV line ($X=50 \Omega$) has $k=0.4$ and operates at $\delta=45^\circ$. Find (a) X_{eff} ; (b) the current; (c) the power; (d) the capacitor voltage; (e) the capacitor rating.

Solution. (a) $X_{eff} = 30 \Omega$.

(b) $I = 2(127.02 \text{ kV}) \sin 22.5^\circ / 30 = 3240.5 \text{ A}$.

(c) $P = 220^2 \sin 45^\circ / 30 = 1140.8 \text{ MW}$.

(d) $V_C = I X_C = 3240.5 \times 20 = 64.81 \text{ kV per phase}$.

(e) $Q_C = 3 I^2 X_C = 630.0 \text{ Mvar}$.

Note. The capacitor voltage of 64.8 kV per phase fixes the platform insulation and the MOV protective level, so it is as important a design result as the Mvar rating.

Example 7.15 Midpoint voltage and power of a compensated line

A symmetric line with $V_s = V_r = 1.0 \text{ pu}$, $X = 0.5 \text{ pu}$ and $k = 0.3$. Find the midpoint voltage, the transmitted power and the uncompensated power at $\delta = 30^\circ$, 60° and 90° ; then (d) the maximum power and its increase.

Solution. $X_{eff} = 0.35 \text{ pu}$.

(a) 30° : $V_m = 0.966 \text{ pu}$, $P = 1.429 \text{ pu}$, uncompensated 1.000 pu.

(b) 60° : $V_m = 0.866 \text{ pu}$, $P = 2.474 \text{ pu}$, uncompensated 1.732 pu.

(c) 90° : $V_m = 0.707 \text{ pu}$, $P = 2.857 \text{ pu}$, uncompensated 2.000 pu.

(d) $P_{max} = 2.857 \text{ pu}$, 42.9% above the uncompensated 2.0 pu. The midpoint voltage, $V \cos(\delta/2)$, is the same with and without the capacitor for the same angle; what changes is the power at that angle.

Note. The midpoint voltage depends only on the angle. Series compensation does not support the midpoint voltage; it lets the line carry more power at the same angle.

Example 7.16 Voltage at the capacitor terminal

Two buses with $V_1 = 1 \angle 30^\circ$ and $V_2 = 1 \angle 0^\circ \text{ pu}$ are connected through a generator reactance $X_g = 0.1 \text{ pu}$ at each end and a line of $X_L = 1.0 \text{ pu}$. A series capacitor is placed next to the sending-end machine. For $X_C = 0, 0.3, 0.4, 0.5, 0.6$ and 0.7 pu find (a) the net reactance; (b) the current; (c) the voltage at the line-side capacitor terminal F; (d) the degree of compensation; (e) the practical conclusion.

Solution. (a) $X_{net} = 1.2 - X_C$.

(b) $I = |V_1 - V_2| / X_{net} = 0.5176 / X_{net}$, at 15° : 0.431, 0.575, 0.647, 0.739, 0.863 and 1.035 pu.

(c) $V_F = V_1 - j I (X_g - X_C)$: 0.990, 1.036, 1.067, 1.114, 1.187 and 1.307 pu.

(d) 0 to 70%.

(e) The current rises by 140% and the capacitor terminal reaches 1.31 pu at 70% compensation. Fault current and insulation set a practical limit of 60% to 70%.

Note. The overvoltage is local to the capacitor terminal: with both bus voltages at 1.0 pu, the line-side terminal of a heavily compensated segment can exceed 1.3 pu.

Example 7.17 SSR frequency screening

A line with total reactance $X = 60 \Omega$ at 50 Hz is compensated 40%. A turbine-generator has a torsional mode at 32 Hz. Find (a) X_C ; (b) f_e ; (c) the rotor frequency $f_0 - f_e$; (d) whether the 32 Hz mode is excited; (e) the compensation that would excite it; (f) the compensation that would excite a 20 Hz mode.

Solution. (a) 24Ω .

(b) $f_e = 50 \sqrt{0.4} = 31.62 \text{ Hz}$.

(c) 18.38 Hz.

(d) No. The rotor frequency (18.38 Hz), not f_e , must be compared with the mode.

(e) $f_e = 50 - 32 = 18$ Hz, so $k = (18/50)^2 = 12.96\%$.

(f) $f_e = 30$ Hz, $k = 36\%$. A 20 Hz mode is therefore critical close to the chosen 40% level.

Note. Screening must use the complementary frequency seen by the rotor, and it must be repeated for every outage that changes the total series reactance of the path.

Example 7.18 Critical clearing angle with series compensation

$P_m = 0.6$ pu is delivered to an infinite bus. Without compensation $P_{max} = 1.0$ pu; with $k = 1/3$ it is 1.5 pu. During the fault $P_{max,f} = 0.4$ pu; the post-fault network equals the pre-fault network. Find (a) δ_0 and δ_{max} without compensation; (b) the critical clearing angle; (c) δ_0 and δ_{max} with compensation; (d) the critical clearing angle; (e) the critical clearing angle if the controlled compensation keeps the fault curve at 0.6 pu.

Solution. The equal-area condition gives
 $\cos \delta_{cr} = [P_m(\delta_{max} - \delta_0) + P_{max} \cos \delta_{max} - P_{max,f} \cos \delta_0] / (P_{max} - P_{max,f})$.

(a) $\delta_0 = 36.87^\circ$, $\delta_{max} = 143.13^\circ$.

(b) $\cos \delta_{cr} = [0.6(1.8546) - 0.8 - 0.32] / 0.6 = -0.0121$; $\delta_{cr} = 90.7^\circ$.

(c) $\delta_0 = 23.58^\circ$, $\delta_{max} = 156.42^\circ$.

(d) $\delta_{cr} = 108.6^\circ$.

(e) 126.4° . A critical clearing angle smaller than the pre-fault angle, as reported in some solutions, signals an algebraic error.

Note. The larger critical clearing angle follows directly from the raised post-fault curve; a controlled device that also lifts the fault-on curve gains a further 18 degrees.

Example 7.19 Series versus shunt support for a 30% increase

A 220 kV line ($X = 50 \Omega$) operates at $\delta = 45^\circ$. Find (a) the compensation for a 30% increase at the same angle; (b) X_C ; (c) the new current; (d) the capacitor rating; (e) the largest increase an ideal midpoint shunt compensator holding 1.0 pu could give at this angle; (f) the midpoint voltage a shunt device would need.

Solution. (a) $k = 1 - 1/1.3 = 0.2308$.

(b) 11.54Ω .

(c) 2527.6 A.

(d) 221.1 Mvar.

(e) $P_{mid}/P = 2 \sin 22.5^\circ / \sin 45^\circ = 1.082$: only 8.2%.

(f) $V_m/V = 1.3 \cos 22.5^\circ = 1.201$ pu, an unacceptable overvoltage. At moderate angles series compensation is far more effective.

Note. To match a 23% series capacitor, a midpoint shunt device would have to hold an unacceptable 1.2 pu voltage, which shows why series compensation is preferred for transfer capability.

Example 7.20 GCSC reactance at several delays

A GCSC has $C = 1500 \mu\text{F}$. Find (a) X_C ; (b) $X_C(\gamma)$ at 20° , 50° and 70° ; (c) the equivalent capacitances; (d) the blocking angles.

Solution. (a) 2.1221Ω .

(b) Equation (7.8): factors 0.5732, 0.1310 and 0.0176, so 1.2163, 0.2779 and 0.0374Ω .

(c) $1/(\omega X)$: 2617, 11453 and $85147 \mu\text{F}$.

(d) $\xi = \pi - 2\gamma$: 140° , 80° , 40° .

Note. The equivalent capacitance becomes very large as the turn-off delay approaches 90 degrees, which is another way of saying that the capacitor is bypassed for almost the whole cycle.

Example 7.21 GCSC voltages, harmonics and THD

A GCSC with $C = 1000 \mu\text{F}$ carries a line current of 20 A peak. For $\gamma = 10^\circ, 25^\circ, 45^\circ$ and 60° find (a) the base voltage $I/\omega C$; (b) the fundamental RMS voltage; (c) the total RMS voltage; (d) the 3rd, 5th and 7th harmonics; (e) the THD; (f) a check of the RMS formula.

Solution. (a) 63.66 V.

(b) Equation (7.7) divided by $\sqrt{2}$: 35.11, 21.53, 8.18 and 2.60 V.

(c) Equation (7.9): 35.33, 22.45, 9.56 and 3.58 V.

(d) 10° : 3.17, 1.75, 1.09 V; 25° : 6.01, 1.89, 0.17 V; 45° : 4.78, 0.96, 0.68 V; 60° : 2.07, 1.24, 0.44 V.

(e) 11.1%, 29.4%, 60.4% and 95.2%.

(f) A numerical RMS of the waveform of Equation (7.6) gives the same values to five figures. With $\sin^2 2\gamma$ in place of $\sin^2 \gamma$ the formula would give 41.2 V instead of 22.45 V at 25° , and its radicand turns negative beyond about 60° ; the two forms coincide only at 60° , which is why a check at that single angle cannot expose the error.

Note. THD rises steeply with the delay because the fundamental falls faster than the harmonic content; the capacitor and valve are rated for the total RMS voltage, not the fundamental.

Example 7.22 GCSC delay for a required reactance

A GCSC with $C = 1000 \mu\text{F}$ must insert 1.5Ω . Find (a) X_C ; (b) the required ratio; (c) γ ; (d) the blocking angle.

Solution. (a) 3.1831Ω .

(b) $X_C(\gamma)/X_C = 0.4712$.

(c) Solving Equation (7.8) numerically: $\gamma = 25.39^\circ$.

(d) $\xi = 129.2^\circ$.

Note. Because the reactance law is nonlinear, the controller stores this inverse relation as a look-up table or solves it numerically at each control step.

Example 7.23 GCSC in voltage compensating mode

A GCSC must maintain 5 kV (peak fundamental) for line currents from 100 A to 400 A peak. Find (a) X_C ; (b) C ; (c) γ at 400 A; (d) the converter-style rating; (e) γ at 100 A.

Solution. (a) $X_C = V_{C_{max}}/I_{min} = 50 \Omega$.

(b) $63.66 \mu\text{F}$.

(c) $400 \times 50 F(\gamma) = 5000$, $F = 0.25$: $\gamma = 39.40^\circ$.

(d) $V_{C_{max}} I_{max}/2 = 1.0 \text{ MVA}$ per phase (peak quantities).

(e) 0° , full insertion.

Note. At maximum current the capacitor is bypassed for most of each half-cycle, so the valve conduction loss is highest exactly where the inserted reactance is smallest.

Example 7.24 GCSC design for a 400 kV line

A 400 kV line ($X = 80 \Omega$) carries 1000 A peak; compensation between 10% and 40% is required. Find (a) X_C and C ; (b) γ for 10%; (c) the rating; (d) the peak capacitor voltage; (e) the transmitted power at 30° for 0, 10% and 40%; (f) the third harmonic at 10%.

Solution. (a) $X_C = 32 \Omega$; $C = 99.47 \mu\text{F}$.

- (b) $F=0.25$; $\gamma=39.40^\circ$.
- (c) $(1000/\sqrt{2})^2 \times 32 = 16.0$ Mvar per phase.
- (d) 32 kV, reached at full insertion ($I X_C$).
- (e) 1000, 1111 and 1667 MW.
- (f) $F_3=12.43\%$ of $I X_C$; 2.81 kV RMS.

Note. At 10% compensation the third harmonic is about half of the fundamental capacitor voltage, which is why single-module phase control is avoided and sequential modular control is used.

Example 7.25 Sequential control of a GCSC

A GCSC of total 3.0Ω is split into $m=4$ modules and carries 400 A peak. Find (a) the module reactance and rating; (b) the reactance obtained with a single module at $\gamma=45^\circ$; (c) the sequential realization of the same reactance; (d) the third harmonic in both cases.

Solution. (a) 0.75Ω ; $(400/\sqrt{2})^2 \times 0.75 = 60$ kvar per module.

- (b) $3.0 \times 0.1817 = 0.545 \Omega$.
- (c) No module fully inserted; one module supplies $0.545/0.75 = 72.7\%$ of its reactance, at $\gamma = 12.49^\circ$.
- (d) Single: 90.0 V RMS. Sequential: 18.1 V RMS, a five-fold reduction. It exceeds the nominal factor $m=4$ because the controlled module sits near full insertion, where its harmonics are small.

Note. The fundamental reactance is the same in both realizations; only the harmonic content and the voltage that each valve must block change.

Example 7.26 TSSC stages and levels

A TSSC has four modules of 5Ω and carries 800 A peak. For 0 to 4 modules inserted, find (a) the reactance; (b) the peak voltage; (c) the reactive power.

Solution. (a) 0, 5, 10, 15, 20 Ω .

- (b) 0, 4, 8, 12, 16 kV.
- (c) $I_{rms}^2 X$: 0, 1.6, 3.2, 4.8, 6.4 Mvar per phase.

Note. The reactive rating grows linearly with the number of inserted stages because every stage carries the same current; the valve current rating does not change.

Example 7.27 TSSC in voltage compensating mode

A TSSC has five 8Ω modules and must hold its rated voltage from 400 A to 1200 A. Find (a) V_{Cmax} ; (b) the stages in service and the voltage at 400, 600, 800, 1000 and 1200 A.

Solution. (a) $V_{Cmax} = m X_C I_{min} = 16$ kV.

- (b) 400 A: 5, 16.0 kV; 600 A: 3, 14.4 kV; 800 A: 2, 12.8 kV; 1000 A: 2, 16.0 kV; 1200 A: 1, 9.6 kV. The stepped approximation leaves gaps of up to 40% at some currents, which is the price of switching.

Note. Equal stages cannot hold the voltage closely over the whole current range; finer or binary-weighted stages narrow the gaps at the cost of more valves.

Example 7.28 TSSC insertion offset and bypass surge

A TSSC stage has $X_C = 12 \Omega$; the line current is 1000 A peak. Find (a) C ; (b) the AC peak capacitor voltage; (c) the maximum voltage after insertion at a current zero; (d) the characteristic impedance with a 0.5 mH valve reactor; (e) the surge current and initial di/dt if bypassed at 8 kV; (f) the remedy.

Solution. (a) 265.3 μ F.

- (b) 12 kV.

- (c) 24 kV, Equation (7.11).
- (d) $Z_0 = \sqrt{L_v/C} = 1.373 \Omega$; ring frequency 437 Hz.
- (e) $I_{surge} = 8000/1.373 = 5.83 \text{ kA}$; $di/dt = V/L = 16 \text{ A}/\mu\text{s}$.
- (f) Bypass at capacitor voltage zero, and size the valve reactor for the valve's di/dt capability.

Note. The offset doubles the peak capacitor voltage after insertion, and a bypass at non-zero voltage produces a surge of several times the peak line current.

Example 7.29 Power sharing between parallel paths with a TSSC

Two parallel paths share 600 MW: line 1 ($X_1 = 60 \Omega$) carries a TSSC of four 10Ω stages, line 2 has $X_2 = 40 \Omega$. Treating the angle across the corridor as adjusted to keep the total constant, find the sharing for 0 to 4 stages inserted.

Solution. Parallel paths share in inverse proportion to reactance: $P_1 = P_t X_2 / (X_1' + X_2)$, with $X_1' = 60 - 10n$.

$n=0$: 240/360 MW; $n=1$: 266.7/333.3; $n=2$: 300/300; $n=3$: 342.9/257.1; $n=4$: 400/200 MW. Two stages equalize the loading, and the full bank shifts 160 MW onto line 1. This is the power-flow-control use of series devices in meshed networks.

Note. Inserting stages draws power into the compensated path; the same principle underlies modular series compensation for congestion management (Section 7.7.1).

Example 7.30 SSR screening of a TSSC

A TSSC of four 10Ω stages is installed on a path whose total reactance, generator and transformer included, is 80Ω . A torsional mode lies at 20 Hz. Find f_e and $f_0 - f_e$ for 1 to 4 stages, and the maximum stages that keep a 5 Hz margin.

Solution. $n=1$: 17.68 / 32.32 Hz; $n=2$: 25.00 / 25.00 Hz; $n=3$: 30.62 / 19.38 Hz; $n=4$: 35.36 / 14.64 Hz. Three stages put the rotor frequency within 0.6 Hz of the mode. Two stages leave 5 Hz margin, and four leave 5.4 Hz but pass through the dangerous setting during switching. The TSSC must either avoid the three-stage state or be combined with an SSR-neutral TCSC module.

Note. The three-stage state lies within 0.6 Hz of the torsional mode, so the controller must block that state, not merely avoid it in steady operation.

Example 7.31 Simplified versus exact TCSC reactance

A TCSC has $C = 1000 \mu\text{F}$ and $L = 1.5 \text{ mH}$. Find (a) X_C , X_L and ω ; (b) the exact resonance; (c) the simplified-model resonance; (d) the reactance at $\alpha = 20^\circ$ by both models; (e) at $\alpha = 50^\circ$; (f) at $\alpha = 60^\circ$; (g) the conclusion.

Solution. (a) $X_C = 3.1831 \Omega$, $X_L = 0.4712 \Omega$, $\omega = 2.599$.

- (b) $\beta_r = 90^\circ / \omega = 34.63^\circ$, that is $\alpha_r = 55.37^\circ$.
- (c) $X_L(\alpha) = X_C$ gives $\alpha_r = 48.21^\circ$.
- (d) $\beta = 70^\circ$: simplified $+1.108 \Omega$, exact $+0.745 \Omega$ (both inductive).
- (e) $\beta = 40^\circ$: simplified -27.60Ω (capacitive), exact $+5.128 \Omega$ (inductive).
- (f) $\beta = 30^\circ$: simplified -5.214Ω , exact -8.111Ω ; the exact value was confirmed by direct time-domain solution.
- (g) Between 48° and 55° the two models disagree on the *sign* of the reactance. The exact characteristic must be used to set the vernier limits.

Note. The two models disagree on the sign of the reactance between about 48 and 55 degrees, so vernier limits derived from the simplified model are unsafe.

Example 7.32 Current division in the TCSC

In the simplified model, the controlled reactor has $X_L(\alpha) = 1.5 X_C$. Find (a) X_{TCSC}/X_C ; (b) the mode; (c) I_{TCR}/I ; (d) I_C/I ; (e) the exact half conduction angle giving the same boost for $X_L/X_C = 0.15$.

Solution. (a) $1.5/(1 - 1.5) = -3$.

(b) Capacitive, boost factor 3.

(c) $I_{TCR}/I = X_C/(X_C - X_L(\alpha)) = -2$: the reactor current is twice the line current and opposite in direction.

(d) $I_C = I - I_{TCR}$, so $|I_C| = 3I$.

(e) Equation (7.16) with $X_L = 0.15 X_C$: $\beta = 30.99^\circ$ ($\alpha = 59.0^\circ$), close to resonance at 34.86° . A boost of 3 is therefore an aggressive operating point.

Note. A boost of 3 leaves only about 4 degrees of margin to resonance for this reactor; practical boost limits are set with a wider margin.

Example 7.33 TCSC design for a 220 kV line

A 220 kV line ($X = 50 \Omega$) carrying 1200 A RMS needs controllable compensation from 20% to 40%. With the TCSC capacitor equal to the minimum compensation ($X_C = 10 \Omega$, blocked mode), find (a) the required boost; (b) C ; (c) the angle for 40% for $X_L/X_C = 0.1, 0.15$ and 0.2 ; (d) L in each case; (e) the resonance margin; (f) the fundamental voltage and rating at maximum compensation; (g) the peak capacitor voltage and thyristor current for $X_L = 1.5 \Omega$.

Solution. (a) $20/10 = 2$.

(b) $318.3 \mu\text{F}$.

(c) Equation (7.16) $\beta = 20 \Omega$: $\beta = 23.90^\circ, 28.53^\circ$ and 32.32° .

(d) $3.18, 4.77$ and 6.37 mH .

(e) Resonance at $28.46^\circ, 34.86^\circ$ and 40.25° : margins of $4.6^\circ, 6.3^\circ$ and 7.9° . The larger reactor gives more room but more harmonics and thyristor current. With $X_L/X_C = 0.1$ ($\omega = 3.16 > 3$) a second resonance also lies at $\beta = 85.4^\circ$, close to full conduction, so the transition to bypass must cross it quickly.

(f) $V_1 = 1200 \times 20 = 24 \text{ kV RMS}$; $Q = I^2 \times 20 = 28.8 \text{ Mvar}$ per phase.

(g) From the exact waveform: peak capacitor voltage 30.6 kV , thyristor RMS current 1707 A and peak 4.24 kA , against a peak line current of 1.70 kA .

Note. The thyristor RMS current exceeds the line current by more than 40%, a direct consequence of the charge-reversal pulses that the simplified model omits.

Example 7.34 TCSC reactance for constant power during a swing

A 500 kV line ($X = 100 \Omega$) with a TCSC at -15Ω operates at $\delta = 30^\circ$. Find (a) the net reactance; (b) the power; (c) the net reactance that holds the power if δ rises to 35° ; (d) the new TCSC setting.

Solution. (a) 85Ω . (b) 1470.6 MW . (c) $X' = V^2 \sin 35^\circ / P = 97.51 \Omega$. (d) -2.49Ω . For power-flow control the TCSC *reduces* its compensation as the angle rises. For damping, the opposite action is needed (more compensation while the angle increases), which is why the two functions are separate loops.

Note. Power-flow control and damping act in opposite directions during a swing, so they are implemented as separate loops with a defined priority.

Example 7.35 SSR and the TCSC

A turbine-generator feeds a line of total reactance 60Ω compensated 45% by a fixed capacitor. Find (a) X_C ; (b) f_e and the rotor frequency; (c) the reactances of a TCSC module ($X_C = 10 \Omega, X_L = 1.333 \Omega$ at 50 Hz) at 33 Hz ; (d) the qualitative effect of replacing part of the fixed bank by that TCSC with capacitor-voltage-synchronized firing.

Solution. (a) 27Ω .

(b) $f_e = 33.54 \text{ Hz}; f_0 - f_e = 16.46 \text{ Hz}.$

(c) $X_C(33) = 15.15 \Omega, X_L(33) = 0.88 \Omega.$

- (d) A plain capacitor stays capacitive at 33 Hz and keeps the resonance. With reversals held at 50 Hz intervals, the TCSC presents an inductive, resistive impedance to the 33 Hz current, which removes that part of the capacitance from the subsynchronous circuit and damps it.

Note. The benefit depends on firing synchronized to the capacitor voltage; a TCSC synchronized to a poorly filtered line-current PLL does not provide it reliably.

Example 7.36 TCSC currents and losses

A TCSC with $X_C = 5 \Omega$, $X_L = 0.667 \Omega$ and valve-reactor resistance 0.05Ω carries 800 A RMS at $\beta = 25^\circ$. Using the exact model, find (a) the resonance angle; (b) X_{TCSC} ; (c) the fundamental voltage; (d) the RMS thyristor and capacitor currents; (e) the losses; (f) the losses as a percentage of the module rating.

Solution. (a) 32.87° .

(b) -8.00Ω (boost 1.6).

(c) 6.40 kV RMS; total RMS 6.44 kV.

(d) $I_{T,rms} = 723 \text{ A}$ (peak 1.92 kA); $I_{C,rms} = 1389 \text{ A}.$

(e) $723^2 \times 0.05 = 26.2 \text{ kW}.$

- (f) 5.12 Mvar rating per phase: losses 0.51%. Using the simplified model, which ignores the large reversal current pulses, would underestimate the reactor current.

Note. Losses of about 0.5% of the module rating are modest, but they rise steeply as the boost approaches resonance.

Example 7.37 TCSC in the admittance matrix

A 400 kV line of 60Ω connects buses k and m with $\theta_k - \theta_m = 10^\circ$ and 1.0 pu voltages; base 100 MVA. A TCSC at -20Ω is added. Find (a) the line reactance in pu; (b) the branch admittance without and with the TCSC; (c) B_{kk} and B_{km} ; (d) the flows; (e) the equivalent power injections.

Solution. (a) $Z_B = 1600 \Omega$; 0.0375 pu, TCSC -0.0125 pu.

(b) $-j 26.67$ pu and $-j 40.0$ pu.

(c) With the TCSC, $B_{kk} = -40$ and $B_{km} = +40$ pu.

(d) $P = \sin 10^\circ / X$: 463.1 MW and 694.6 MW.

- (e) Keeping the original admittance, the TCSC is replaced by injections $P_k = -231.5$ MW at bus k and $+231.5$ MW at bus m (with corresponding reactive injections). This power-injection form leaves the admittance matrix constant during iterations.

Note. The power-injection form is convenient in Newton-Raphson iterations because the admittance matrix stays constant while the TCSC setting changes.

Example 7.38 SSSC power and equivalent capacitor

A two-machine system has $V_s = V_r = 1$ pu, $X = 1$ pu and an SSSC with $V_q = 0.6$ pu at $\delta = 20^\circ$. Find (a) the uncompensated power; (b) the power with the SSSC; (c) the current; (d) the capacitor giving the same power at this angle; (e) the power at $\delta = 0$.

Solution. (a) 0.342 pu. (b) Equation (7.20): $0.342 + 0.6 \cos 10^\circ = 0.933$ pu. (c) Equation (7.19): 0.947 pu. (d) $X_C = V_q / I = 0.633$ pu, $k = 63\%$. (e) 0.6 pu with the SSSC, but zero with any capacitor.

Note. The SSSC contribution is largest at small angles, exactly where a series capacitor contributes least.

Example 7.39 SSSC range and power reversal

$V = 1$ pu, $X = 1.2$ pu, $\delta = 40^\circ$, $V_{qmax} = 0.5$ pu. Find (a) the uncompensated power; (b) the maximum power; (c) the minimum power; (d) the percentage control range; (e) the angle below which power reverses at $V_q = -0.5$ pu.

Solution. (a) 0.536 pu. (b) 0.927 pu. (c) 0.144 pu. (d) +73 % and -73 %. (e) $2 \sin(\delta/2) < 0.5$: $\delta < 28.96^\circ$.

Note. Reversing the sign of the injected voltage reverses the flow at small angles, which no passive series element can do.

Example 7.40 SSSC on a 400 kV line

A 400 kV line ($X = 100 \Omega$) operates at 45° ; the SSSC injects up to 30 kV (line-to-line). Find (a) P_0 ; (b) the SSSC term; (c) the power range; (d) the current with full capacitive injection; (e) the SSSC MVA rating.

Solution. (a) 1131.4 MW. (b) $400 \times 30 \times \cos 22.5^\circ / 100 = 110.9$ MW, not 1108.7 MW. (c) 1020.5 to 1242.2 MW. (d) 1940.7 A (against 1767.5 A without injection). (e) $3(30/\sqrt{3})(1.941) = 100.8$ MVA.

Note. With line-to-line kilovolts and ohms the SSSC term is already in megawatts; no further unit factor is needed.

Example 7.41 SSSC with storage compensating R and X

A 275 kV line has $X = 75 \Omega$, $R = 10 \Omega$ and operates at $\delta = 35^\circ$. An SSSC with storage must raise power by 25% while keeping $X/R = 7.5$. Find (a) the base power, reactive power and current; (b) the target power; (c) R_{eff} and X_{eff} ; (d) the new current; (e) the voltage components; (f) the MVA rating and the power drawn from storage.

Solution. (a) From $S = 3 V_r I^*$: $P_0 = 544.4$ MW, $|Q_0| = 254.9$ Mvar, $I_0 = 1262$ A.

(b) 680.5 MW.

(c) Solving with $X_{eff} = 7.5 R_{eff}$: $R_{eff} = 8.0 \Omega$, $X_{eff} = 60 \Omega$.

(d) 1577 A.

(e) $V_q = (75 - 60) \times 1577 = 23.66$ kV per phase; $V_p = (10 - 8) \times 1577 = 3.15$ kV per phase.

(f) $3 \times 23.87 \times 1.577 = 113.0$ MVA; the storage supplies $3 V_p I = 14.9$ MW, which is the share of the line's $I^2 R$ loss now carried by the SSSC.

Note. The 14.9 MW supplied by the store is continuous, so it must come from a real energy source rather than from the DC capacitor.

Example 7.42 SSSC in impedance compensating mode

A 500 kV line of 200Ω is to behave as 150Ω ; $V_{qmax} = 80$ kV line-to-line. Find (a) the reactance to emulate; (b) the maximum current for which this holds; (c) the MVA rating; (d) the current at $\delta = 30^\circ$; (e) the largest angle at which the emulation holds.

Solution. (a) 50Ω . (b) 46.19 kV/ $50 = 924$ A. (c) 128 MVA. (d) 996 A, above the limit, so the SSSC saturates in voltage mode. (e) $\delta = 2 \sin^{-1}(I_{max} X_{eff} / 2 V_{ph}) = 27.8^\circ$.

Note. Beyond the current limit the emulated reactance can no longer be held, and the SSSC continues along its voltage limit.

Example 7.43 Capacitor versus SSSC for a 50% increase

A 110 kV line ($X = 40 \Omega$) operates at 45° . Compare a fixed capacitor and an SSSC that each raise power by 50% at this angle: (a) P_0 ; (b) the capacitor reactance; (c) its rating; (d) the SSSC voltage; (e) its rating; (f) the behaviour at other angles.

Solution. (a) 213.9 MW. (b) $k = 1/3$, $X_C = 13.33 \Omega$. (c) $I = 1823$ A, 132.9 Mvar. (d) $V_q = 106.95 / (2.75 \cos 22.5^\circ) = 42.1$ kV line-to-line. (e) 132.9 MVA, identical at this operating point because current and voltage are the same. (f) At smaller angles the capacitor contributes less and the SSSC relatively more.

The SSSC can also reduce power, but its rating is a converter rating, typically several times dearer per MVA than capacitor kvar. A hybrid of the two often gives the best economy.

Note. At this one operating point the two solutions need the same rating, but the SSSC also offers bidirectional control and freedom from classical series resonance.

Numerical consistency note. All SSSC examples use three-phase quantities. With line-to-line RMS voltage and line current the converter rating is $S = \sqrt{3} V_{LL} I$; with per-phase voltage it is $S = 3 V_{ph} I$. Per-unit calculations use one declared three-phase base and need no additional $\sqrt{3}$.

Unsolved Practice Problems

P7.1 A 345 kV line of 100Ω operates at 35° . Find the degree of compensation for a 40% increase in power, the capacitor reactance, the new current and the three-phase capacitor rating.

P7.2 A path of 70Ω total reactance is compensated 35%. Find X_C , f_e and the rotor frequency, and the compensation at which a 25 Hz torsional mode would be excited.

P7.3 With $P_m = 0.7$ pu, $P_{max} = 1.2$ pu (uncompensated) or 1.6 pu (compensated) and a fault curve of 0.3 pu, find the initial angle and critical clearing angle in both cases.

P7.4 A GCSC has $C = 800 \mu\text{F}$. Find X_C , the reactance at $\gamma = 30^\circ$ and 60° , and the delay for 1.0Ω .

P7.5 A GCSC with $C = 600 \mu\text{F}$ carries 300 A peak at $\gamma = 40^\circ$. Find the fundamental RMS voltage, the total RMS voltage, the third harmonic RMS voltage and the THD.

P7.6 A GCSC must hold 8 kV (peak) from 200 A to 600 A peak. Find X_C , C and γ at 600 A.

P7.7 A TSSC has six 4Ω stages and carries 1000 A RMS. Find the maximum voltage and reactive power. In voltage compensating mode with $V_{Cmax} = 18$ kV, find the stages in service at 750, 900 and 1500 A.

P7.8 A TSSC stage ($X_C = 10 \Omega$) carries 1500 A peak. Find the maximum voltage after insertion at a current zero with a discharged capacitor, and the surge current if it is bypassed at 6 kV through 0.3 mH .

P7.9 A TCSC has $X_C = 12 \Omega$ and $X_L = 1.6 \Omega$. Find the resonance (β_r and α_r), the exact reactance and boost at $\beta = 20^\circ$, and the exact reactance at $\beta = 60^\circ$.

P7.10 For the TCSC of Problem P7.9, compare simplified and exact reactance at $\beta = 60^\circ$ ($\alpha = 30^\circ$).

P7.11 A line of 60Ω needs 15% to 30% controllable compensation. With $X_C = 9 \Omega$ and $X_L/X_C = 0.15$, find the angle for 30%, C , L and the resonance angle.

P7.12 An SSSC on a line with $V = 1$ pu, $X = 0.9$ pu, $\delta = 30^\circ$ injects ± 0.3 pu. Find the power range and the currents.

P7.13 A 500 kV line of 150Ω at 40° has an SSSC of ± 50 kV (line-to-line). Find the power range.

P7.14 An SSSC must make a 1.0 pu line behave as 0.75 pu with $V_{qmax} = 0.3$ pu. Find the current limit of the emulation and the rating at 1.2 pu current.

P7.15 A 400 kV, 60Ω line between buses at 10° apart receives a TCSC of -12Ω . Find the flows before and after (100 MVA base) and the percentage increase.

P7.16 Compare a TSSC, a TCSC and an SSSC for a meshed 400 kV network where flows must be both raised and lowered, a nearby thermal plant has a 22 Hz torsional mode, and the site is space-limited.

Answer Key (Unsolved Problems)

P7.1 $k = 0.2857$; 28.57Ω ; 1677 A; 241.1 Mvar.

P7.2 24.5Ω ; 29.58 Hz; 20.42 Hz; $k = 25\%$.

P7.3 Uncompensated 35.69° , 83.06° ; compensated 25.94° , 96.33° .

P7.4 $3.979\ \Omega$; $1.556\ \Omega$; $0.229\ \Omega$; 39.30° .

P7.5 272.4 V; 306.6 V; 138.0 V; 51.6%.

P7.6 $40\ \Omega$; $79.58\ \mu\text{F}$; 33.59° .

P7.7 24 kV; 24 Mvar per phase; 6, 5 and 3 stages (18 kV each).

P7.8 30 kV; $Z_0 = 0.971\ \Omega$, 6.18 kA.

P7.9 $\beta_r = 32.86^\circ$, $\alpha_r = 57.14^\circ$; $-14.51\ \Omega$ (boost 1.21); $+2.74\ \Omega$.

P7.10 Simplified $+6.21\ \Omega$; exact $+2.74\ \Omega$.

P7.11 $\beta = 28.53^\circ$ ($\alpha = 61.47^\circ$); $353.7\ \mu\text{F}$; 4.30 mH; $\beta_r = 34.86^\circ$.

P7.12 0.556 pu uncompensated; 0.878 to 0.234 pu; 0.908 and 0.242 pu.

P7.13 1071.3 MW; 914.7 to 1227.9 MW.

P7.14 1.2 pu; 0.36 pu.

P7.15 463.1 MW; 578.8 MW; 25%.

P7.16 Open-ended. The SSSC suits bidirectional control and has no resonance; a TCSC with capacitor-voltage-synchronized firing is SSR-neutral, but it can reduce flow only within its limited inductive vernier; a TSSC is unsuitable near the 22 Hz mode.

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